

1. ELECTRIC CHARGE AND FIELD

Electric Charge -

The property of material by which it exerts or experience electric and magnetic effects is called the electric charge.

Examples -

- (i) Experience of seeing a spark or hearing a crackle, when we take off our synthetic cloths.
- (ii) Lightning in the sky during thunderstorms due to electric discharge.

Electrostatics -

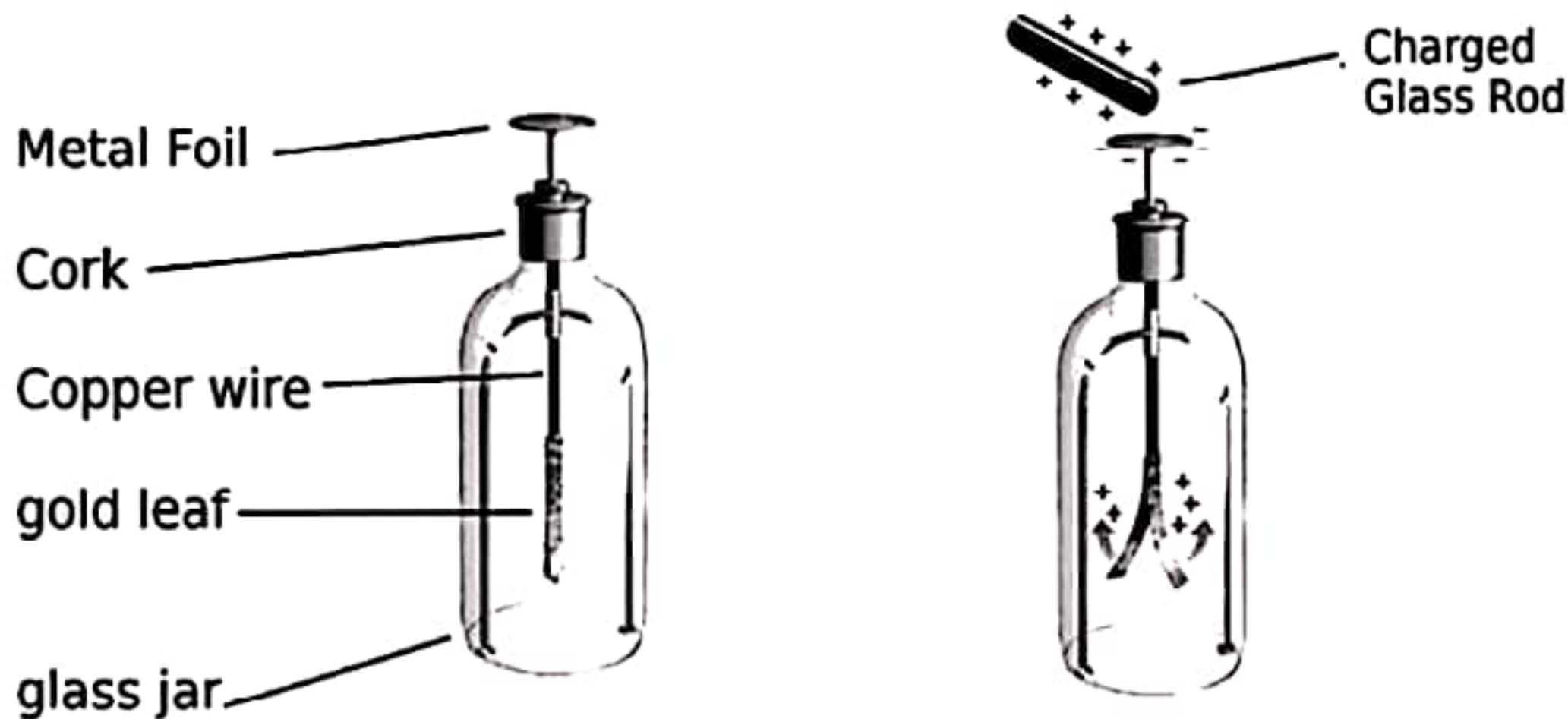
The branch of Physics, which deals with the study of charges at rest (static charges), the forces, fields and potential due to these charges is called Electrostatics or static Electricity.

Important Facts about charge -

- (1.) There are only two types of electric charge.
- (2.) Like charges repel and unlike charges attract each other when they come in contact.
- (3.) The unlike charges nullify (cancel) each other's effect when they come in contact. Therefore the charges are named as positive and negative by the American scientist 'Benjamin Franklin'.
- (4.) Normally the materials are electrically neutral, they do not contain charge because their charges (Protons and electrons) are exactly balanced.
- (5.) The electron of the outermost orbit of an atom are far from the nucleus so these electrons are loosely bounded with the nucleus and can be separated easily from the orbit by giving some energy. These electrons are then called free electrons.

(6.) The body or atom can be charged positively by losing some of its electrons. Similarly a body or atom can be charged negatively by gaining electrons.

(7.) The apparatus used to detect charge on a body is "Gold Leaf Electroscope".



(8.) An electric charge can create electric field (E), magnetic field ($\vec{B}$) and electromagnetic radiations

$$V = 0$$

$\oplus \rightarrow$ Only Electric field ($\vec{E}$)

$$V = \text{Const.}$$

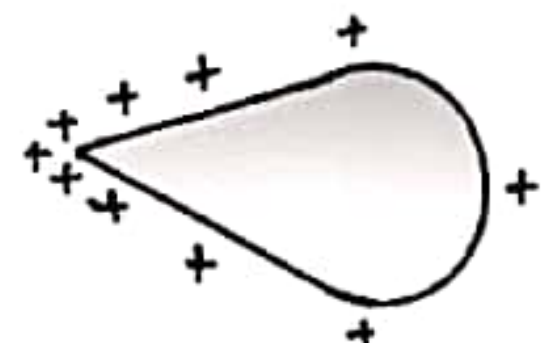
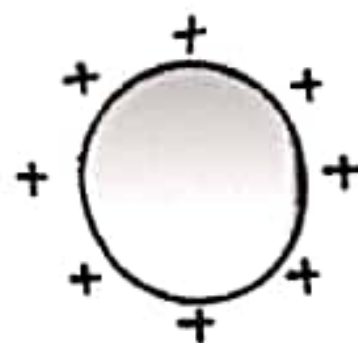
$\oplus \rightarrow$ Electric Field ($\vec{E}$) & Magnetic Field ($\vec{B}$)

$$V \neq \text{Const.}$$

$\oplus \rightarrow \vec{E}, \vec{B}$ and Electromagnetic Radiations

(9.) Electric charge uniformly distribute on a uniform surface but the charge density on a non-uniform surface is maximum on that points where the radius of curve is minimum.

Charge density $\sigma \propto \frac{1}{R}$



Point Charge -

It is a point where all the charge of a body is concentrated.

Conductor, Insulator & Semiconductor -

Conductor -

The material through which the electric charge and current can flow easily are called Conductors.

Examples - All Metals, Human body, earth etc.

Insulator -

The materials which resist the flow of electric charge and current are called insulators.

Examples - Wood, Porcelene, plastic etc.

Semiconductors -

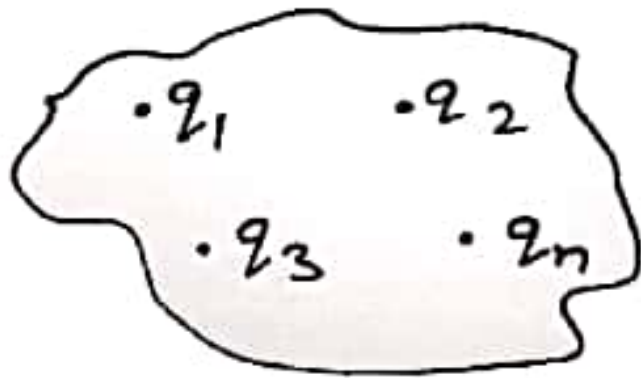
The materials which resist the flow of electric charge and current but the value of their resistance is in between the conductor and insulator are called semiconductors.

Examples - Silicon (Si), Germanium (Ge)
Carbon (C) → Excellent thermal conductivity.

Properties of Electric Charge -

(1) Additivity of Charges:-

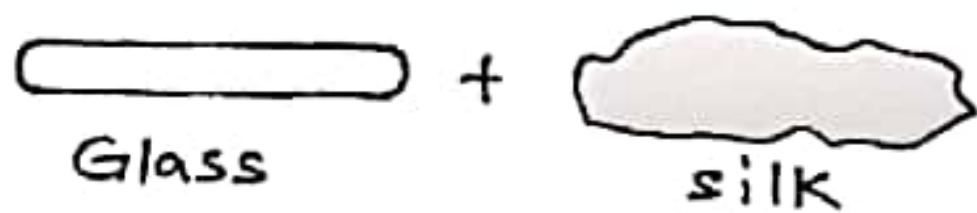
Electric charge is a scalar quantity so the total charge of the system is obtained simply by adding the different charges algebraically with proper sign according to their charge.



$$Q = q_1 + q_2 + \dots + q_n$$

(2) Charge is conserved:-

Charge can not be either created or destroyed. It can only be transferred from one body to another. Hence the total charge of the isolated system is always conserved.



$$\text{Total charge} = 0 + 0 = 0$$



$$\text{Total charge} = +q - q = 0$$

(3) Quantisation of Charge:-

The electric charge on a body is always an integral multiple of e (charge on 1 electron). This is called quantisation of charge.

$$Q = ne$$

$$n = 0, 1, 2, \dots, \infty$$

$$e = 1.602 \times 10^{-19} \text{ C}$$

* The unit of electric charge is coulomb.
1 Coulomb -

The total charge on 6.24×10^{18} electrons is equal to 1 coulomb.

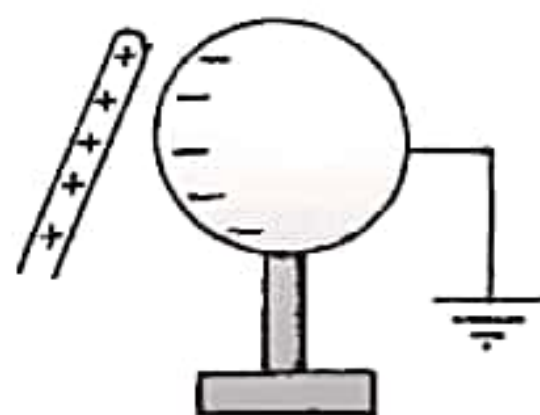
Charging by Induction -

Method 1. (Using a ground Connection)

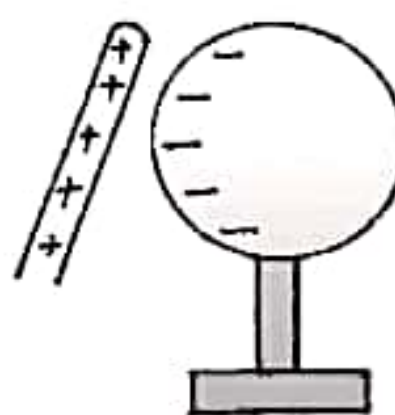
- (i) A positively charged rod is brought near a neutral metal sphere and polarising it.
- (ii) The sphere is grounded allowing electrons to be attracted from the earth.



(a)



(b)



(c)

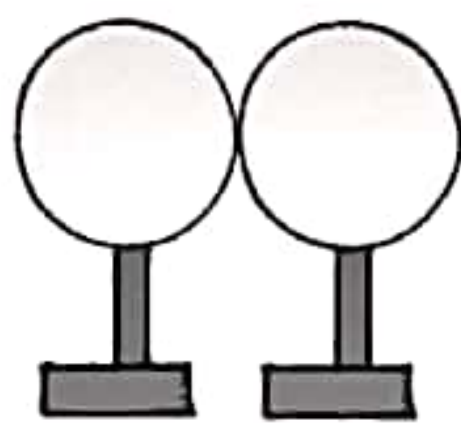


(d)

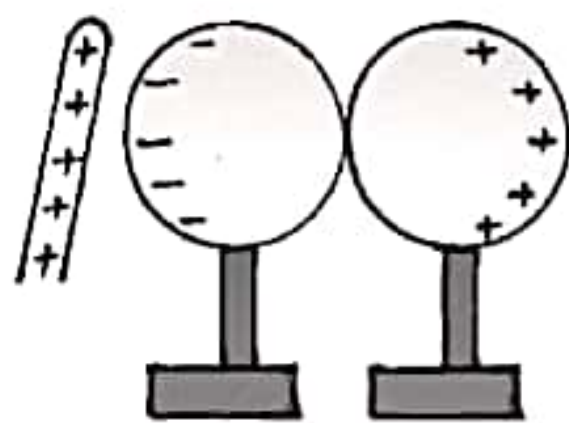
- (iii) The ground Connection is broken.
- (iv) The positive rod is removed leaving the sphere with an induced negative charge.

Method 2.

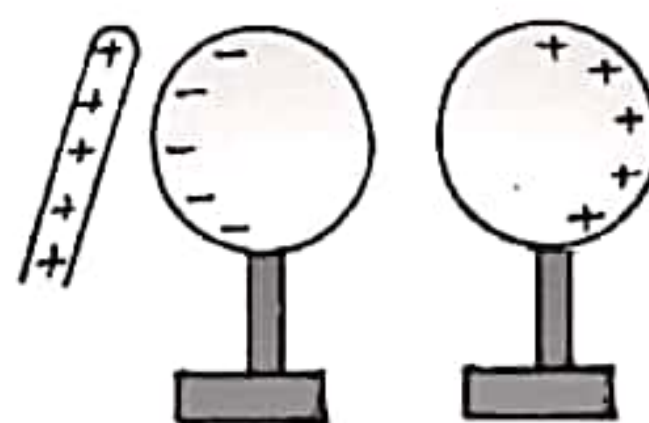
- (i) Two uncharged metal spheres are in contact with each other but insulated from rest of the world.
- (ii) A positively charged rod is brought near the sphere, attracting negative charge and leaving the other sphere positively charged.



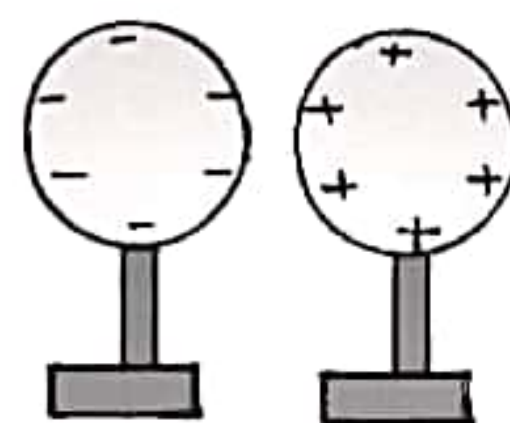
(a)



(b)



(c)



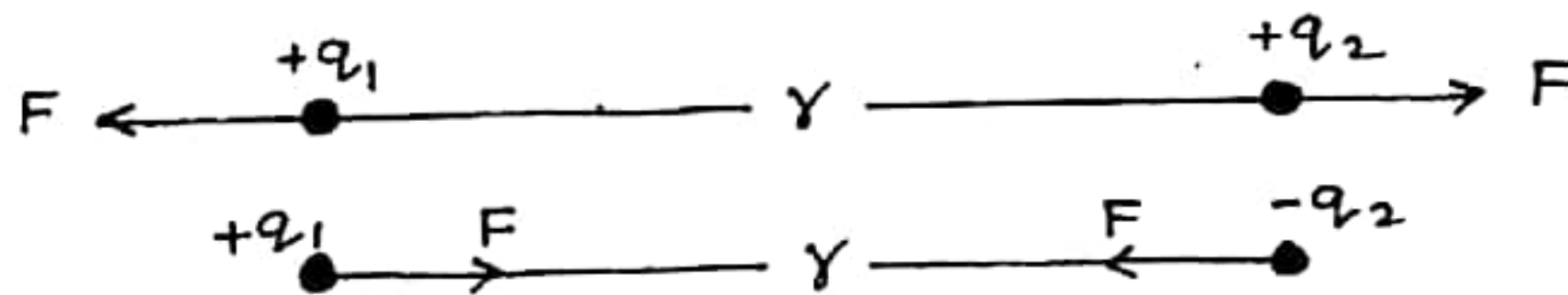
(d)

- (iii) Now the spheres are separated before the rod is removed and thus separating positive & negative charge.
- (iv) Remove the rod. The charges on spheres will rearrange themselves and uniformly distributed over the spheres.

Coulomb's Law-

The force of interaction between any two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them, and acts along the line joining the two charges.

Suppose two point charges q_1 and q_2 are separated in vacuum by a distance r .



According to Coulomb's Law,

$$F \propto q_1 q_2 \quad \text{and} \quad F \propto \frac{1}{r^2}$$

OR

$$F = \frac{K q_1 q_2}{r^2}$$

K - Electrostatic Constant

* Electrostatic constant K depends on the nature of medium separating the charges and on the system of units.

In cgs system, $K = 1$

In SI system, $K = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$

* ϵ_0 electrical permittivity of free space

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$$

Coulomb's Law in vector form -

To represent the direction of force between two point charges, Coulomb's Law is represented in vector form.

Let the position vector of charges q_1 and q_2 be $\vec{r}_1$ and $\vec{r}_2$ respectively.

We denote the force on q_1 due to q_2 by $\vec{F}_{12}$ and the force on q_2 due to q_1 by $\vec{F}_{21}$.

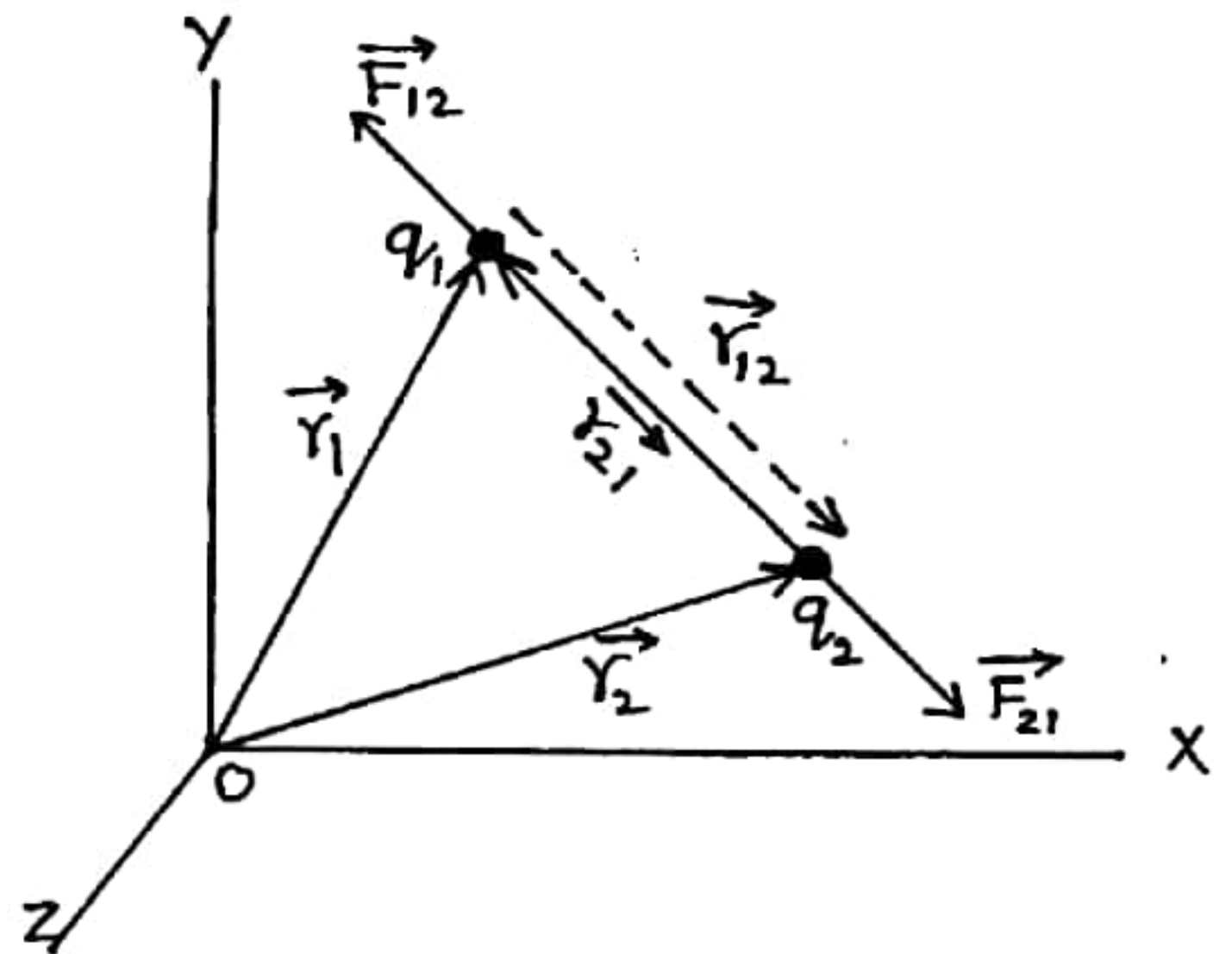
$$\vec{F}_{12} = \vec{F}_{21}$$

Here $\vec{r}_1 + \vec{r}_{12} = \vec{r}_2$

$$\vec{r}_{12} = \vec{r}_2 - \vec{r}_1$$

also $\vec{r}_2 + \vec{r}_{21} = \vec{r}_1$

or $\vec{r}_{21} = \vec{r}_1 - \vec{r}_2$ and $\vec{r}_{12} = -\vec{r}_{21}$



The magnitude of vectors $\vec{r}_{12}$ and $\vec{r}_{21}$ is denoted by $|\vec{r}_{12}|$ and $|\vec{r}_{21}|$ and their unit vectors are given as -

$$\hat{r}_{12} = \frac{\vec{r}_{12}}{|\vec{r}_{12}|} \quad \text{and} \quad \hat{r}_{21} = \frac{\vec{r}_{21}}{|\vec{r}_{21}|}$$

So the force $\vec{F}_{12}$ can be given as -

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\vec{r}_{21}|^2} \hat{r}_{21}$$



- (i) If q_1 and q_2 are of same sign ; $q_1 q_2 > 0$ then $\vec{F}_{12}$ is along $\vec{r}_{21}$, which denotes repulsion
- (ii) If q_1 and q_2 are of opposite sign ; $q_1 q_2 < 0$ then $\vec{F}_{12}$ is along $-\vec{r}_{21}$ or $\vec{r}_{12}$, which denotes attraction

Q. Two point charges of $+2 \mu\text{C}$ and $+6 \mu\text{C}$ repel each other with a force of 12 N . Now if each is given an additional charge of $-4 \mu\text{C}$, what will be the new force?

Sol. Here $q_1 = +2 \mu\text{C}$, $q_2 = +6 \mu\text{C}$, $F = 12 \text{ N}$
and $q'_1 = +2 - 4 = -2 \mu\text{C}$, $q'_2 = +6 - 4 = 2 \mu\text{C}$

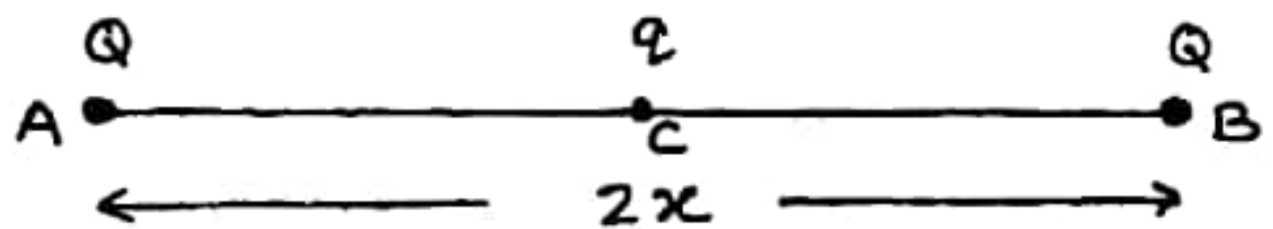
$$\text{So } \frac{F'}{F} = \frac{q'_1 q'_2}{q_1 q_2} = \frac{(-2)(2)}{(2)(6)} = -\frac{1}{3}$$

$$F' = -\frac{F}{3} = -\frac{12}{3} = -4 \text{ N (Attractive)}$$

Q. A charge q is placed at the centre of the line joining two equal charges Q . Show that the system of three charges will be in equilibrium if $q = -Q/4$.

Sol.

Let two equal charges Q each, be held at A and B , where $AB = 2x$.



Here C is the centre of AB , where charge q is held. Net force on q is zero. So q is already in equilibrium.

For the three charges to be in equilibrium, net force on each charge must be zero.

Now, total force on Q at B is -

$$\frac{K Q q}{x^2} + \frac{K Q \cdot Q}{(2x)^2} = 0$$

$$\frac{K Q q}{x^2} = -\frac{K Q^2}{4x^2}$$

so $q = -\frac{Q}{4}$

Hence Proved.

Dielectric Constant (K) -

$$\text{force in a medium (F}_m\text{)} = \frac{1}{4\pi\epsilon} \times \frac{q_1 q_2}{r^2} \quad \text{--- (1)}$$

where ϵ - electrical permittivity of the medium.

$$\text{force in vacuum (F}_0\text{)} = \frac{1}{4\pi\epsilon_0} \times \frac{q_1 q_2}{r^2} \quad \text{--- (2)}$$

Dividing eq. (2) by (1), we get -

$$\boxed{\frac{F_0}{F_m} = \frac{\epsilon}{\epsilon_0} = \epsilon_r \text{ or } K}$$

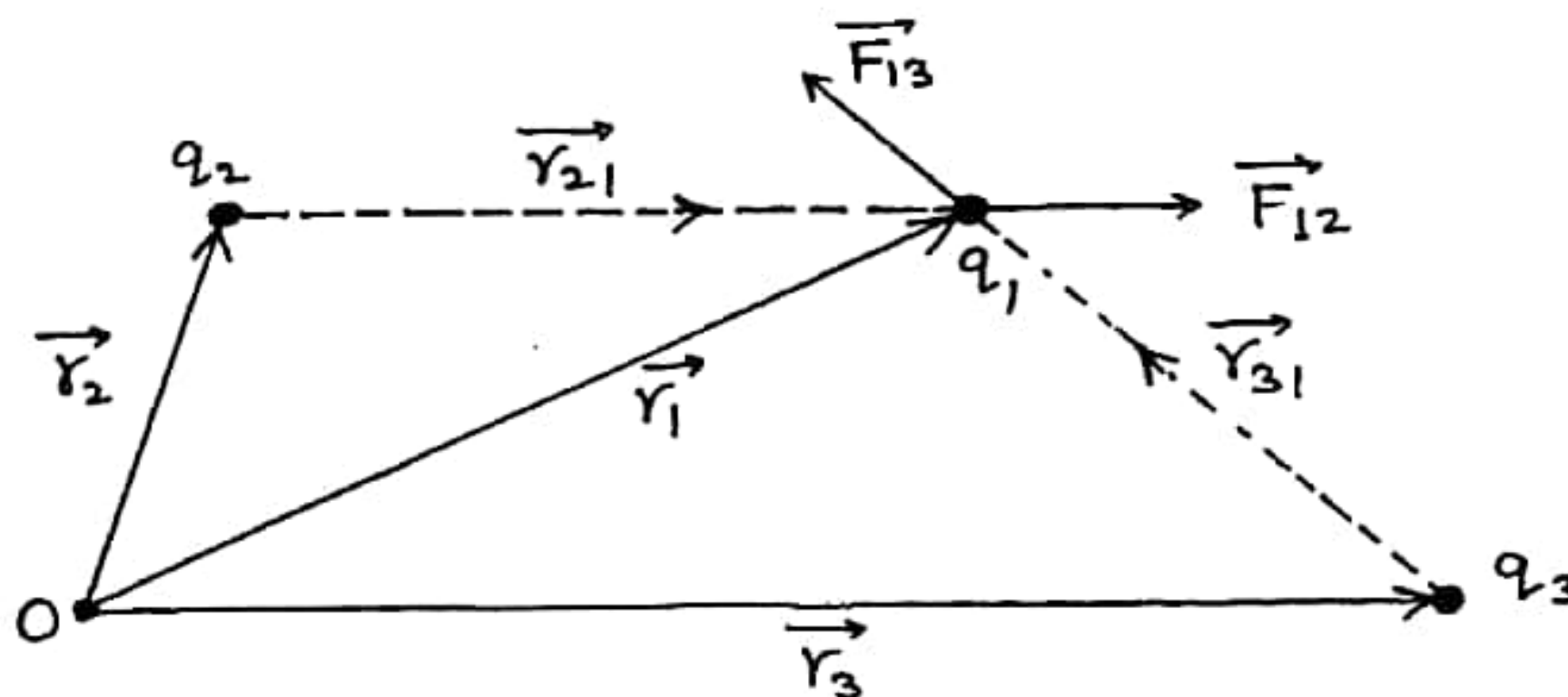
where ϵ_r or K is called dielectric constant or relative electrical permittivity of the medium.

- * Dielectric constant of a medium is the ratio of absolute electrical permittivity of the medium to the electrical permittivity of free space.
- * Thus force between two given charges in a medium (K) is only $\frac{1}{K}$ times of the force between them in air/vacuum.

$$\boxed{F_m = \frac{F_0}{K}}$$

Force between Multiple Charges -

Total force on any charge due to a number of charges at rest is the vector sum of all the forces on that charge due to other charges.



Here $\vec{F}_{12} = \frac{K q_1 q_2}{|\vec{r}_{21}|^2} \hat{r}_{21}$ and $\vec{F}_{13} = \frac{K q_1 q_3}{|\vec{r}_{31}|^2} \hat{r}_{31}$

Thus the resultant force $\vec{F}_1$ on charge q_1 is obtained by

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13}$$

Similarly if a system have n charges $q_1, q_2, \dots, q_n$ then the force on q_1 due to $q_2, q_3, \dots, q_n$ charges can be calculated by the vector sum of forces $\vec{F}_{12}, \vec{F}_{13}, \dots, \vec{F}_{1n}$ as -

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \dots + \vec{F}_{1n}$$

or
$$\vec{F}_1 = K q_1 \sum_{i=2}^n \frac{q_i}{|\vec{r}_{i1}|^2} \hat{r}_{i1}$$

Q. Two equal positive charges each of $2 \mu\text{C}$ interact with a third positive charge of $3 \mu\text{C}$ situated as shown in fig. Calculate the magnitude and direction of the force on the $3 \mu\text{C}$ charge.

Sol.

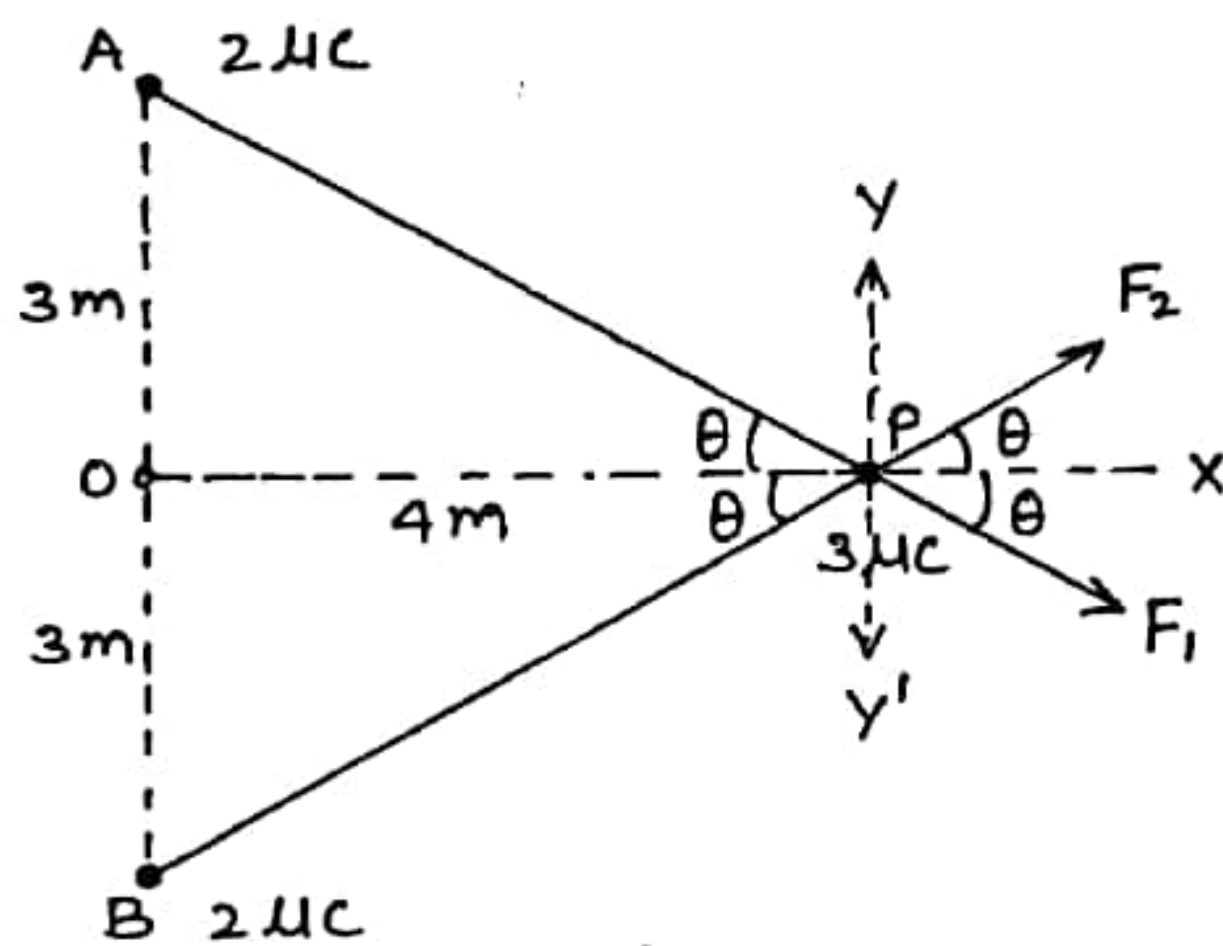
Here $OA = OB = 3 \text{ m}$
and $OP = 4 \text{ m}$

So $AP = BP = \sqrt{3^2 + 4^2} = 5 \text{ m}.$

According to Coulomb's Law Force on charge at P due to charge at A and B are equal in magnitude so-

$$F_1 = F_2 = \frac{K q_1 q_2}{r^2} = \frac{9 \times 10^9 \times 2 \times 10^{-6} \times 3 \times 10^{-6}}{5^2}$$

So $F_1 = F_2 = 2.16 \times 10^{-3} \text{ N}$



Here the components of F_1 and F_2 along Y and Y' axis are equal in magnitude ($F_1 \sin \theta = F_2 \sin \theta$) but opposite in direction so they cancel each other

The Components along PX add up. So total force on charge $3 \mu C$ at P is -

$$F = 2 F_1 \cos \theta = 2 \times 2.16 \times 10^{-3} \times \frac{4}{5}$$

So $F = 3.5 \times 10^{-3} \text{ N}$ — Ans.

Q. Two fixed point charges $+4e$ and $+e$ units are separated by a distance 'a'. where should the third point charge be placed for it to be in equilibrium.

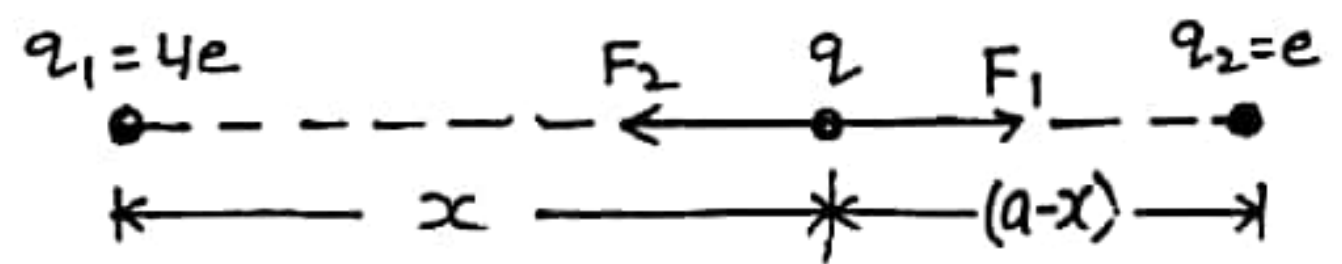
Sol.

For the charge $+q$ to be in equilibrium

$$F_1 = F_2$$

$$\frac{kq \cdot 4e}{x^2} = \frac{kq \cdot e}{(a-x)^2} \Rightarrow \frac{4}{x^2} = \frac{1}{(a-x)^2}$$

on solving - $x = 2a/3$ — Ans.

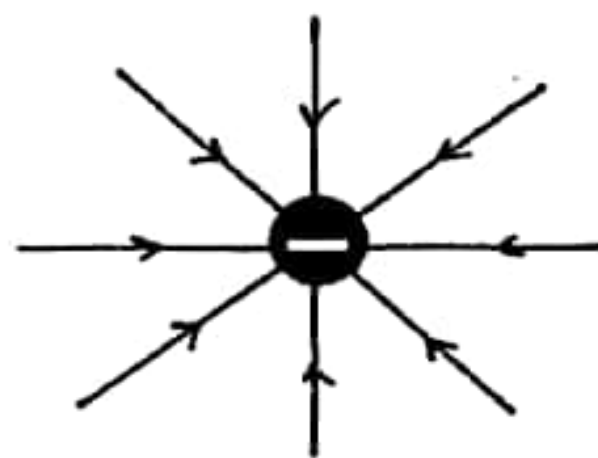
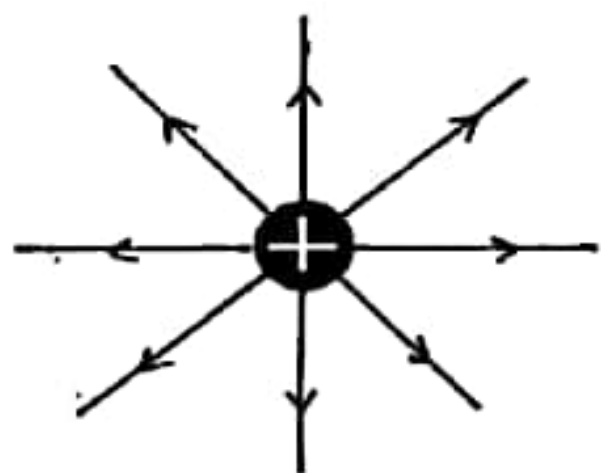


Electric Field -

An electric field is the region of space surrounding electrically charged particles where a force is exerted on other electrically charged objects.

* The direction of the field is taken to be the direction of the force it would exert on a positive test charge.

* The electric field is radially outward from a positive charge and radially inward towards a negative charge.



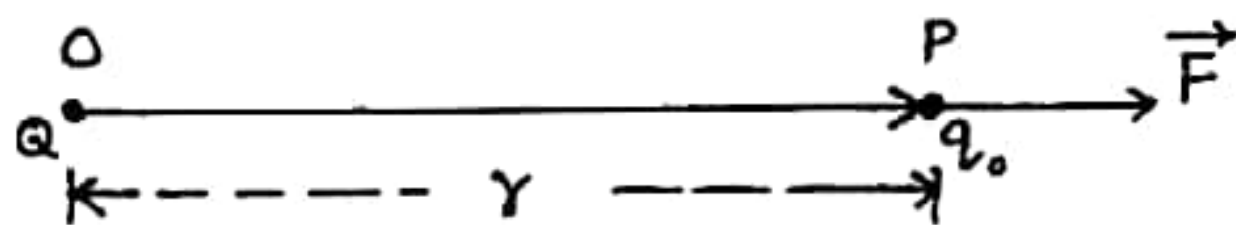
* Concept of Field was introduced first of all by "Faraday".

Electric Field Intensity ($\vec{E}$)

The electric field intensity at any point is the strength of electric field at that point. It is defined as the force experienced by unit Positive charge placed at that Point.

According to Coulomb's Law force on test charge q_0 due to a point charge Q is

$$F = \frac{1}{4\pi\epsilon_0} \frac{Q \cdot q_0}{r^2}$$



by definition

$$\vec{E} = \frac{\vec{F}}{q_0}$$

so

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{r^2} \hat{r}$$

where $\hat{r}$ is unit vector directed from Q towards q_0 .

$$\text{SI Unit of } E = \frac{N}{C}$$

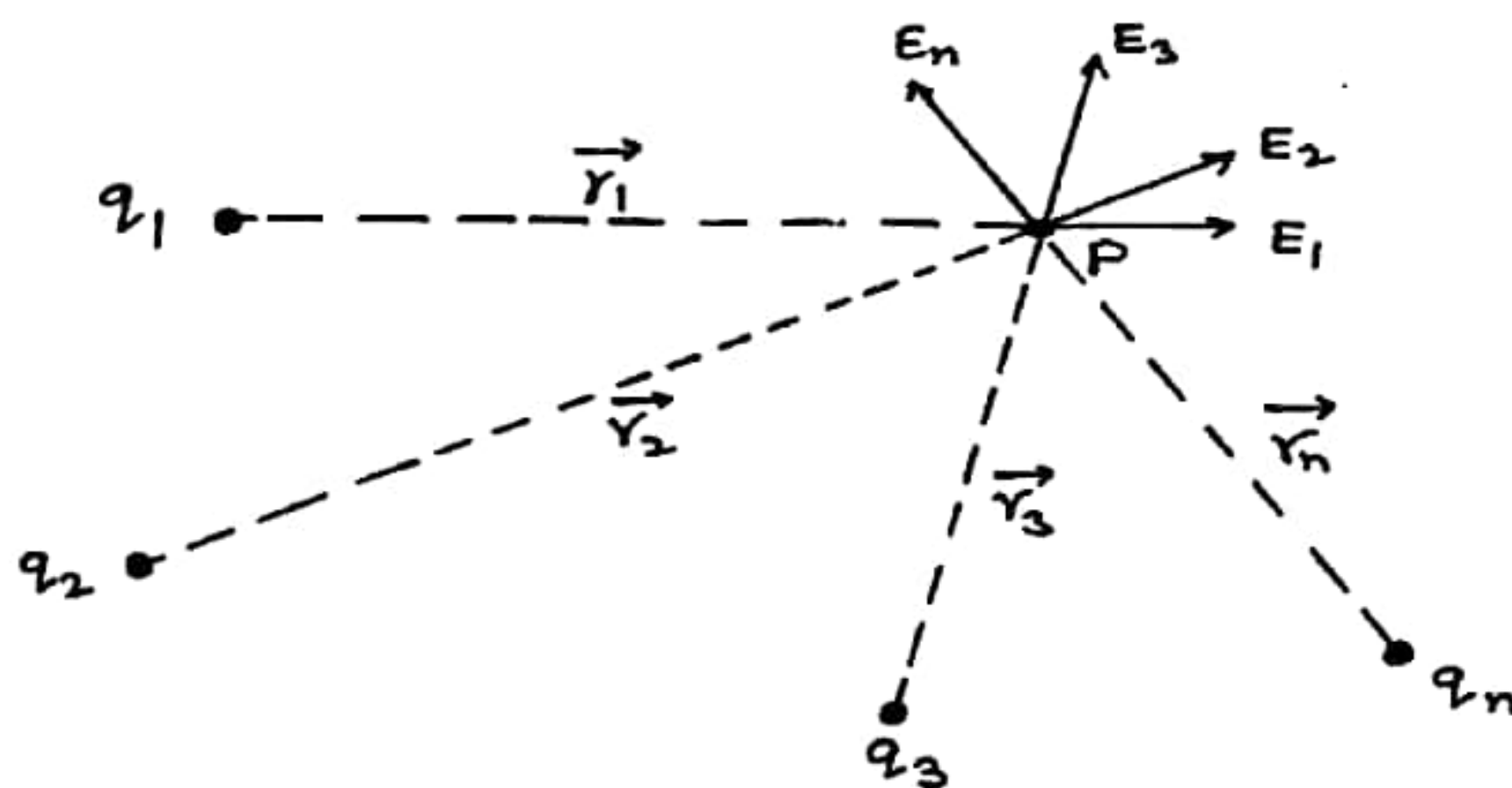
Important Points about Electric field -

- (1) The magnitude of electric field $\vec{E}$ is same at equal distances from the charge Q , thus it has a spherical symmetry.
- (2) The magnitude of the electric field $\vec{E}$ depends only on the charge Q and the distance r .
- (3) The test charge q_0 may have its own electric field, it may modify the electric field of the source charge. Therefore to minimize this effect we rewrite electric field intensity at $\vec{r}$ as.

$$\vec{E} = \lim_{q_0 \rightarrow 0} \frac{\vec{F}}{q_0}$$

Electric Field due to a system of charges -

Consider a system of charges $q_1, q_2, \dots, q_n$ with position vectors $\vec{r}_1, \vec{r}_2, \dots, \vec{r}_n$.



The electric field intensity at a point P due to a system of charges is the vector sum of the electric fields at the point due to individual charges.

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \dots + \vec{E}_n \quad \vec{E} = \frac{Kq_1}{|\vec{r}_1|^2} \hat{r}_1 + \frac{Kq_2}{|\vec{r}_2|^2} \hat{r}_2 + \dots + \frac{Kq_n}{|\vec{r}_n|^2} \hat{r}_n$$

or

$$\vec{E} = K \sum_{i=1}^n \frac{q_i}{|\vec{r}_i|^2} \hat{r}_i$$

Q. What are the magnitude and direction of the electric field at centre of the square in figure.
If $q = 10^{-8} \text{ C}$ and $a = 5 \text{ cm}$.

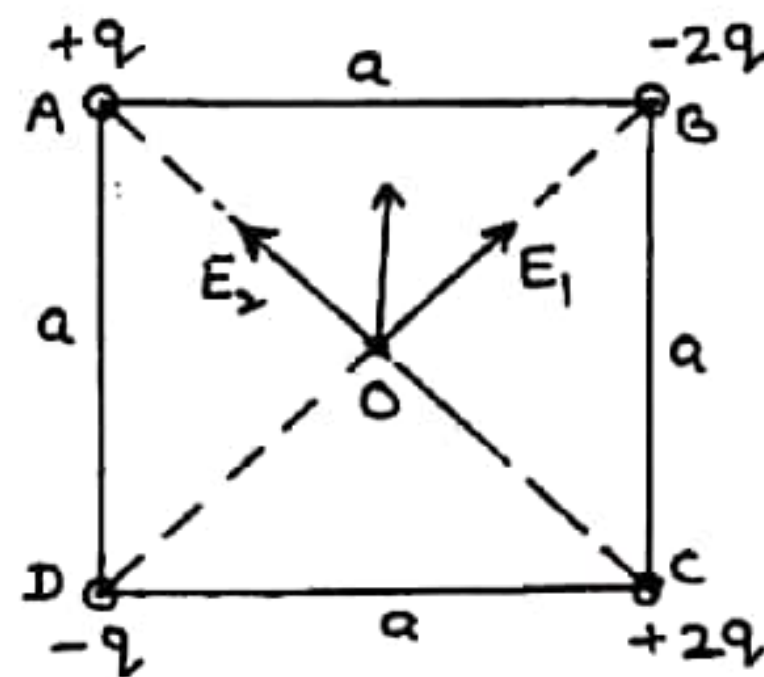
Sol. Here

$$OA = OB = OC = OD = r$$

$$r = \frac{5 \times 10^{-2}}{\sqrt{2}} \text{ m.}$$

also here $E_1 = E_2 = \frac{q}{4\pi\epsilon_0 r^2}$

$$\text{or } E_1 = E_2 = \frac{10^{-8} \times 9 \times 10^9}{(5 \times 10^{-2} / \sqrt{2})^2} = 7.2 \times 10^4 \text{ N/C}$$



In square ABCD the $\angle AOB = 90^\circ$

$$\text{So the Net Electric field at O} = \sqrt{E_1^2 + E_2^2 + 2E_1E_2\cos 90^\circ}$$

$$E = \sqrt{E_1^2 + E_2^2} = E_1\sqrt{2}$$

$$E = 7.2\sqrt{2} \times 10^4 \text{ N/C at } 45^\circ \text{ to OA.}$$

Electric Field Lines -

An electric field line is a curve drawn in such a way that the tangent to it at each point is in the direction of the net field at that point.



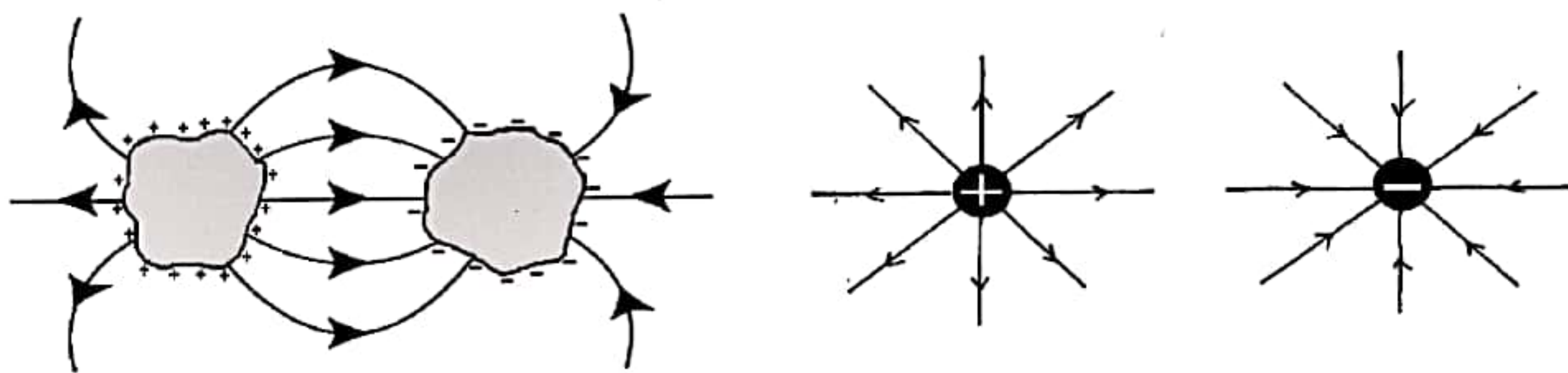
The magnitude of the field is represented by the density of field lines.

Hence Electric field intensity at a point is equal to number of field lines crossing normally a unit area around that point.

Properties of Electric Field Lines -

(1) Electric field lines are continuous curves. They start from a positively charged body and end at a negatively charged body.

If there is a single charge then the field lines may start or end at infinity.

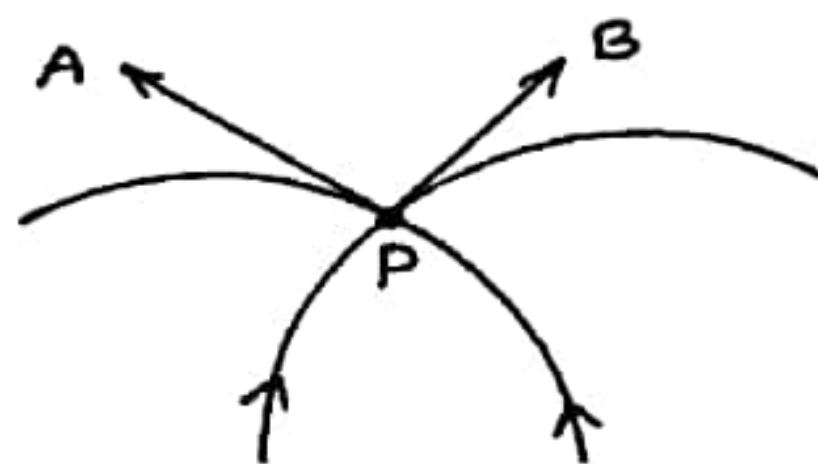


(2) No electric field lines exist inside the charged body. Thus electrostatic field lines do not form continuous closed loops.

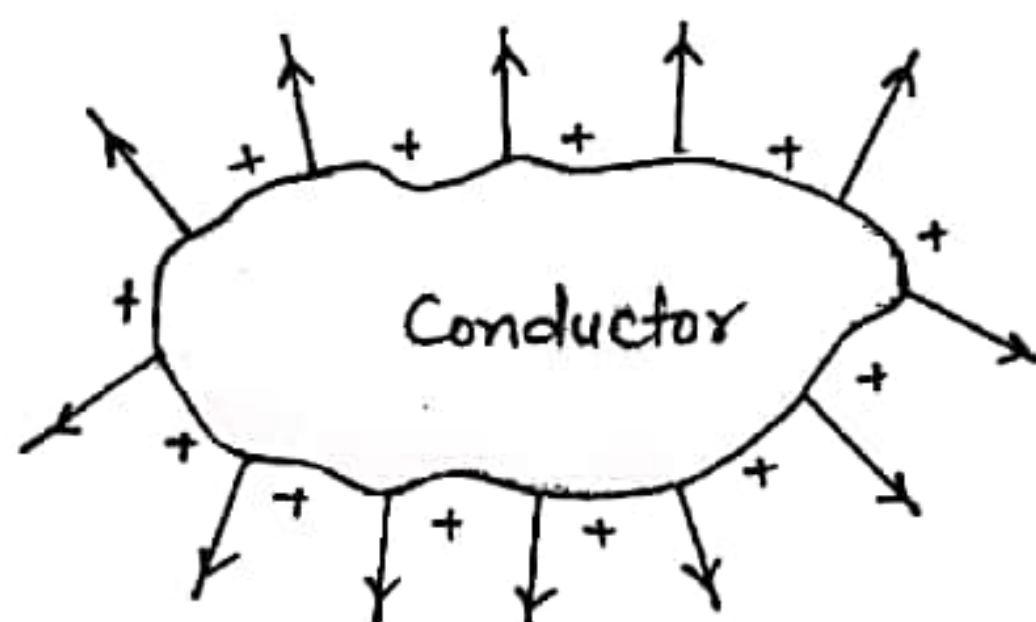
★ The magnetic field lines are continuous (or endless) closed loops as against the electric field lines.

(3) Tangent to the electric field line at any point gives the direction of electric field intensity at that point.

(4) No two electric field lines of force can intersect each other. This is because at the point of intersection P , we can draw two tangents PA and PB . This would mean two directions of electric field intensity at the same point, which is not possible.

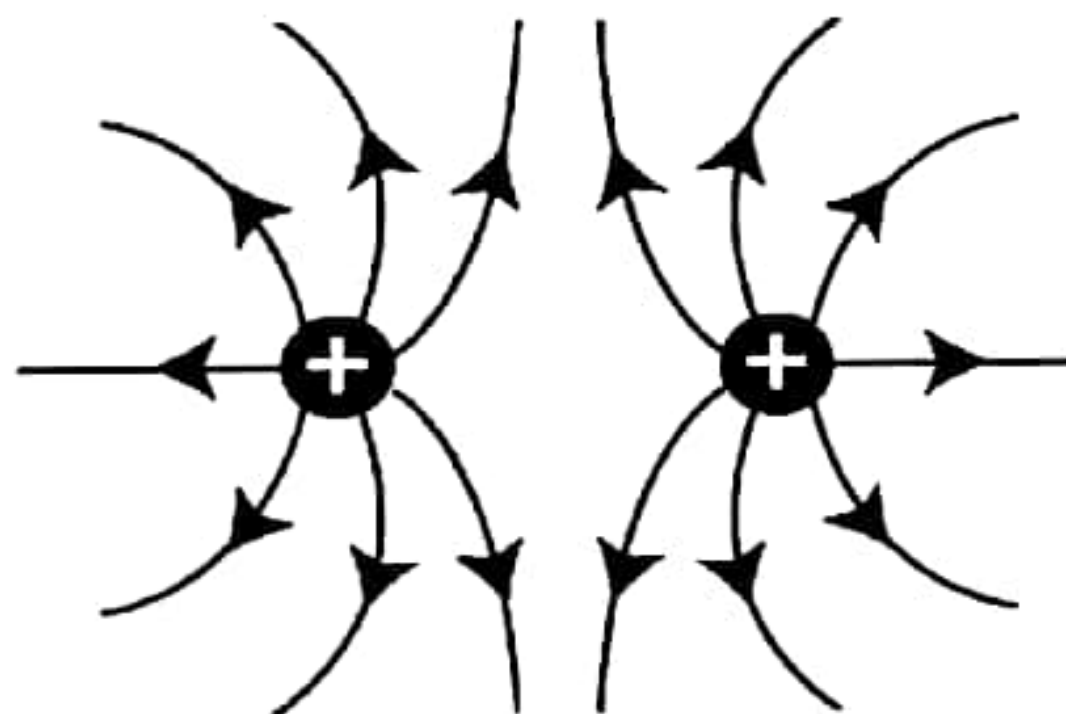
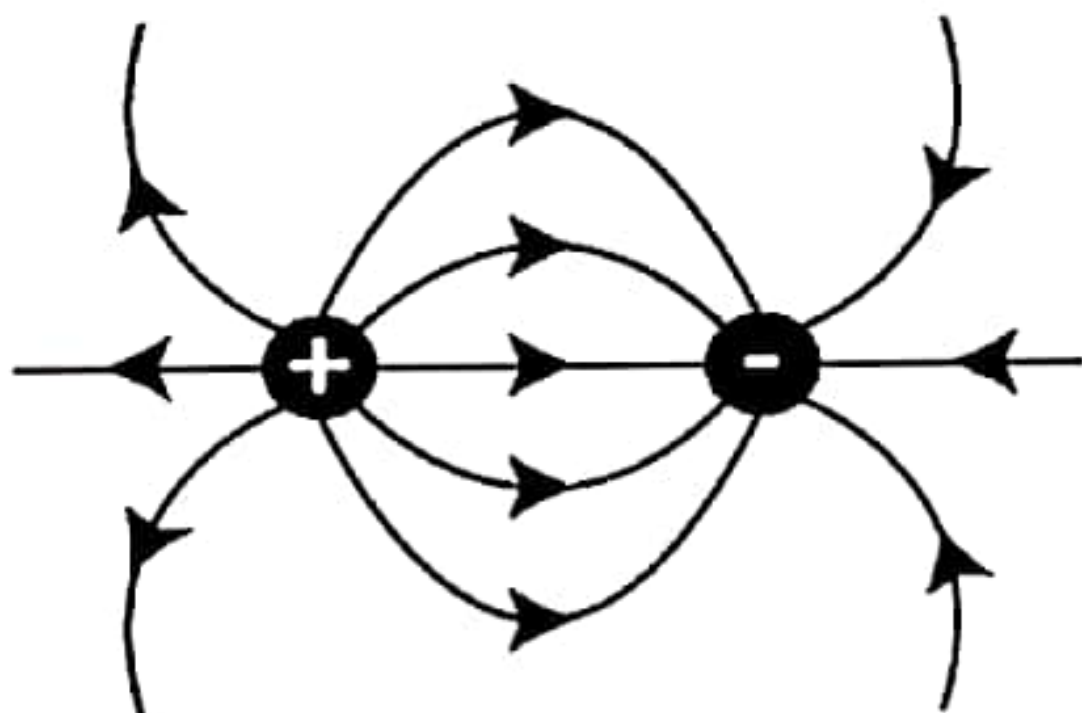


(5) The electric field lines are always normal to the surface of a conductor. So there is no component of electric field intensity parallel to the surface of the conductor.



(6) The field lines around a system of two positive charges (q, q) gives a pictorial representation of their mutual repulsion.

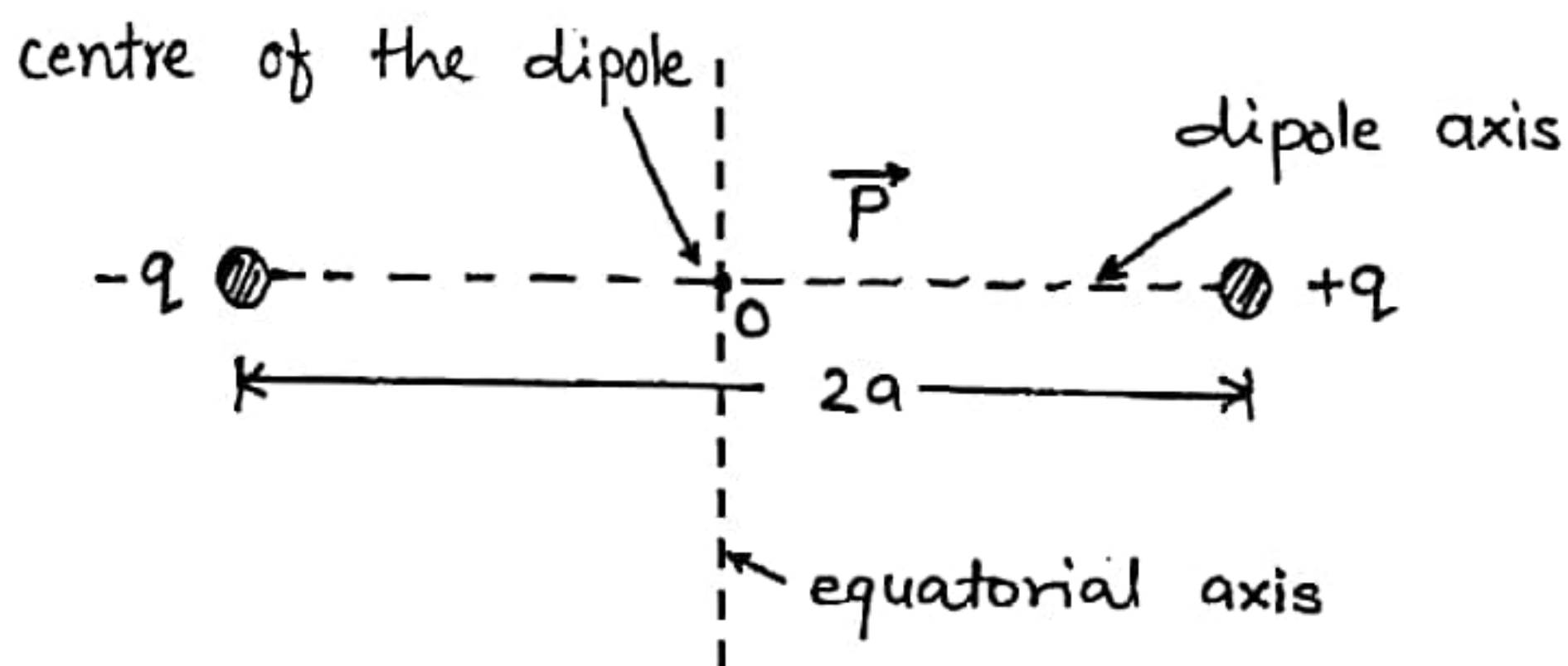
While around the configuration of two equal and opposite charges ($q, -q$), a dipole, shows the mutual attraction between the charges.



Electric Dipole -

An electric dipole is a pair of equal and opposite point charges separated by a short distance.

Let q and $-q$ are two equal and opposite point charges separated by a small distance ' $2a$ '.



* The total charge of the electric dipole is zero, but this does not mean that the field of the electric dipole is zero, because the charge q and $-q$ are separated by some distance, so the electric field due to them when added do not exactly cancel out.

* But at distances much larger than the separation of the two charges forming a dipole ($r \gg 2a$) then the field due to q and $-q$ nearly cancel out.

Dipole Moment ($\vec{P}$) -

Dipole moment is a measure of the strength of electric dipole. It is a vector quantity whose direction is from negative charge to positive charge.

Dipole moment is equal to the product of the magnitude of either charge and the distance between them.

$$\vec{P} = q \times 2a \hat{p}$$

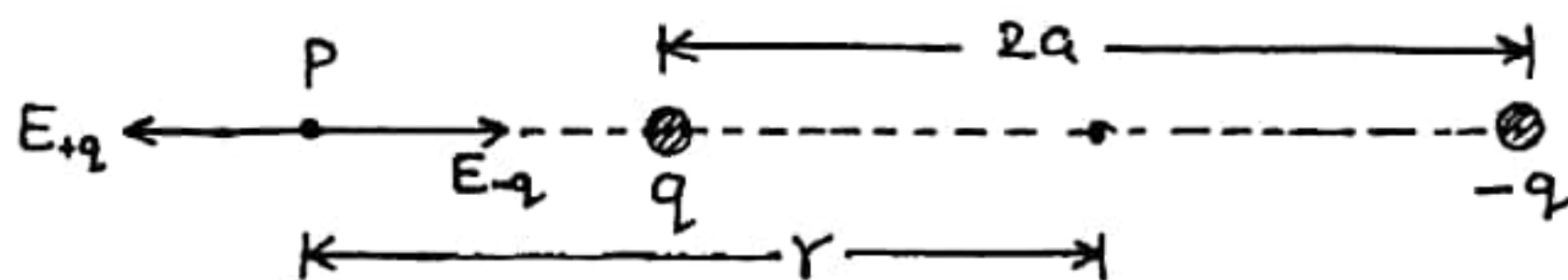
Dipole Field -

It is the space around the dipole in which the electric effect of the dipole can be experienced.

The electric field at any point P is obtained by adding the electric fields E_{-q} due to charge $-q$ and E_{+q} due to the charge q by the parallelogram law of vectors.

(i) Field Intensity on Axial Line of Electric Dipole -

Let the point P be at distance r from the centre of the dipole on the side of the charge q , then



$$\text{Here } E_{-q} = \frac{Kq}{(r+a)^2} \quad \text{and} \quad E_{+q} = \frac{Kq}{(r-a)^2}$$

So the total Electric field at point P is -

$$E = E_{+q} - E_{-q} \quad (E_{+q} > E_{-q})$$

$$E = Kq \left[\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2} \right]$$

$$E = Kq \cdot \frac{4ar}{(r^2 - a^2)^2}$$

For $(r \gg a)$

$$\vec{E} = \frac{4Kqa}{r^3} \hat{p} = \frac{2K\vec{p}}{r^3}$$

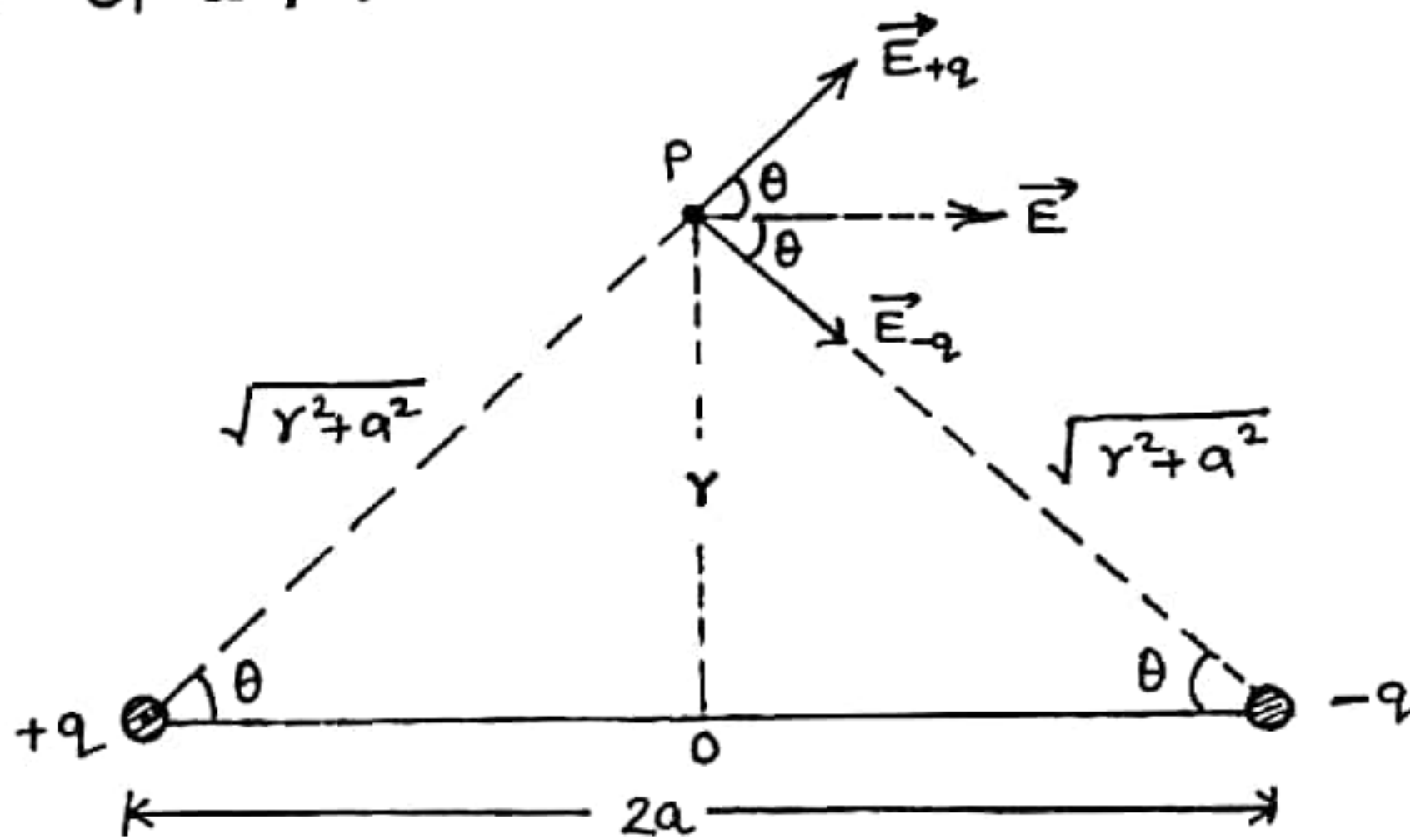
The direction of $\vec{E}$ is along $\vec{p}$ i.e from $-q$ to $+q$.

*

$$E \propto \frac{1}{r^3}$$

(ii) Field Intensity on Equatorial Line of Electric Dipole -

Let the point P is on the equatorial line, where $OP = r$.



So the magnitudes of the electric fields due to the two charges q and $-q$ are given as -

$$E_{+q} = \frac{Kq}{(r^2 + a^2)} \quad \text{and} \quad E_{-q} = \frac{Kq}{(r^2 + a^2)}$$

Here the components of E_{+q} and E_{-q} normal to the dipole axis cancel each other and the components along the dipole axis add up.

So the total electric field is in opposite direction to $\hat{P}$.

$$E = (E_{+q} + E_{-q}) \cos \theta$$

$$E = \frac{K \cdot 2qa}{(r^2 + a^2)^{3/2}}$$

For $(r \gg a)$

$$\vec{E} = -\frac{K \cdot 2qa}{r^3} \hat{P} = -\frac{K \vec{P}}{r^3}$$



* Hence the dipole field at large distances falls off not as $1/r^2$ but as $1/r^3$.

* Due to an electric dipole

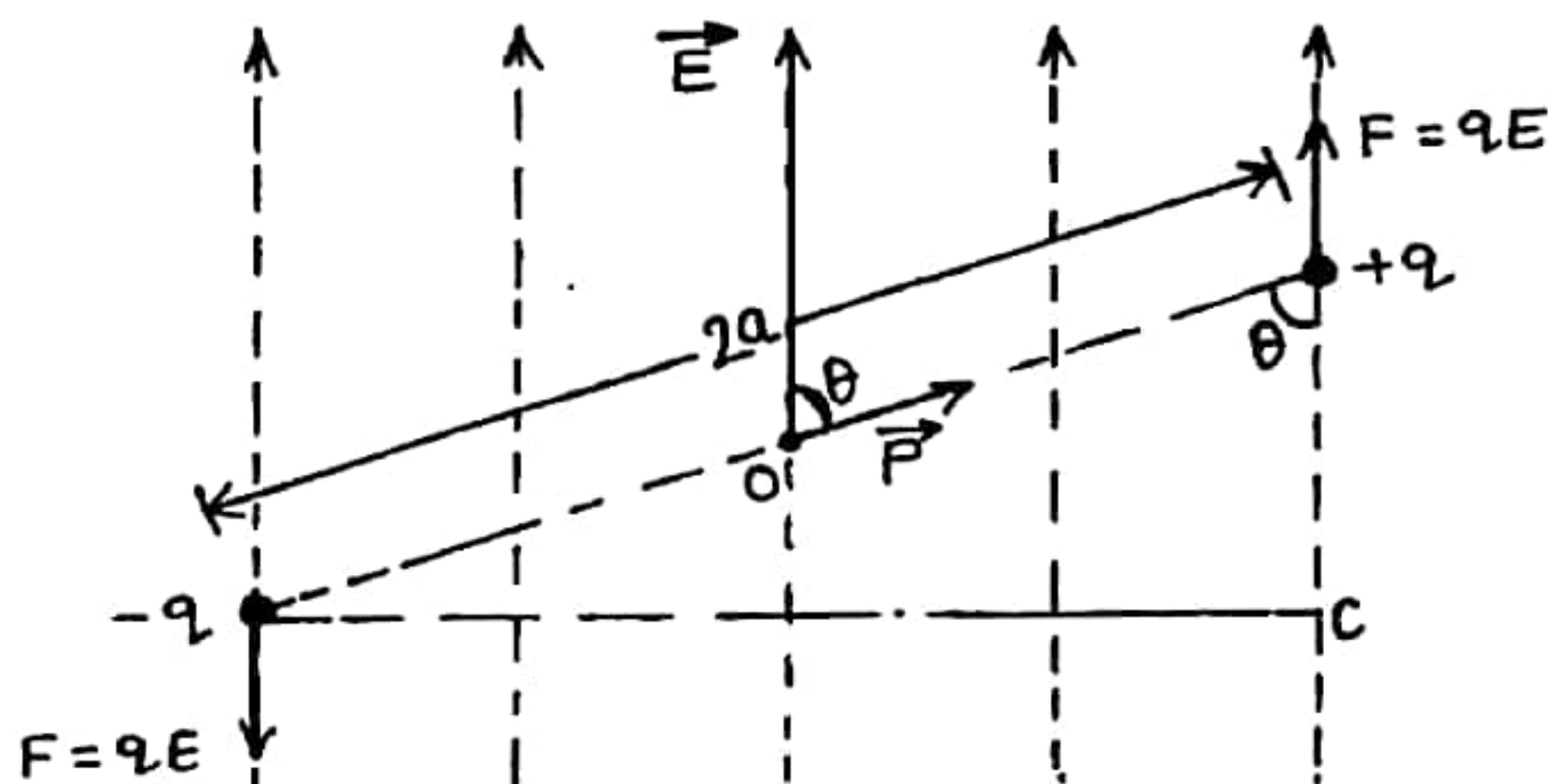
$$E_{\text{axial}} = 2 E_{\text{equatorial}}$$

Dipole in a uniform Electric Field -

Let a dipole be held in a uniform external electric field $\vec{E}$ at an angle θ with the direction of $\vec{E}$.

Force on charge $+q = qE$, along the direction of $\vec{E}$.

Force on charge $-q = qE$, along opposite direction to $\vec{E}$.



Net force on the dipole = zero

The forces on charge $+q$ and $-q$ resulting in a torque on the dipole which rotates the dipole in the anticlockwise direction.

Thus the torque tends to align the dipole axis along the direction of field $\vec{E}$.

The magnitude of torque is equal to the magnitude of each force (qE) multiplied by the perpendicular distance between the two antiparallel forces.

$$\text{Torque } \tau = qE \times 2a \sin \theta$$

$$\tau = pE \sin \theta$$

or

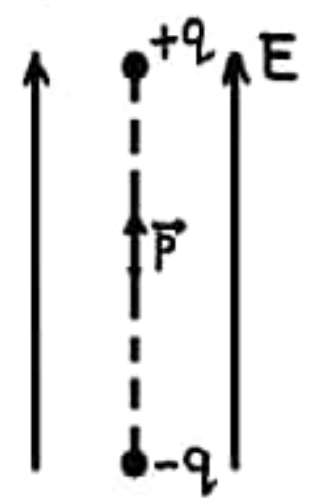
$$\boxed{\vec{\tau} = \vec{p} \times \vec{E}}$$

The direction of $\vec{\tau}$ is given by right handed screw rule and is perpendicular to $\vec{p}$ and $\vec{E}$.

* When $\vec{P}$ is along $\vec{E}$, $\theta = 0^\circ$

$$\tau = PE \sin 0^\circ = 0$$

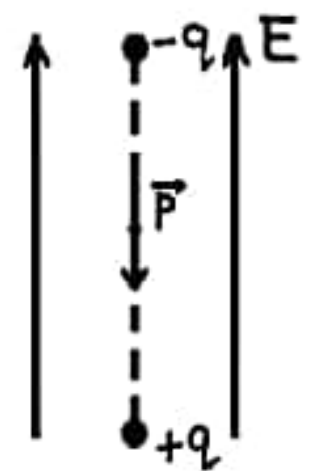
The dipole is in stable equilibrium.



* When $\vec{P}$ is along $\vec{E}$, $\theta = 180^\circ$

$$\tau = PE \sin 180^\circ = 0$$

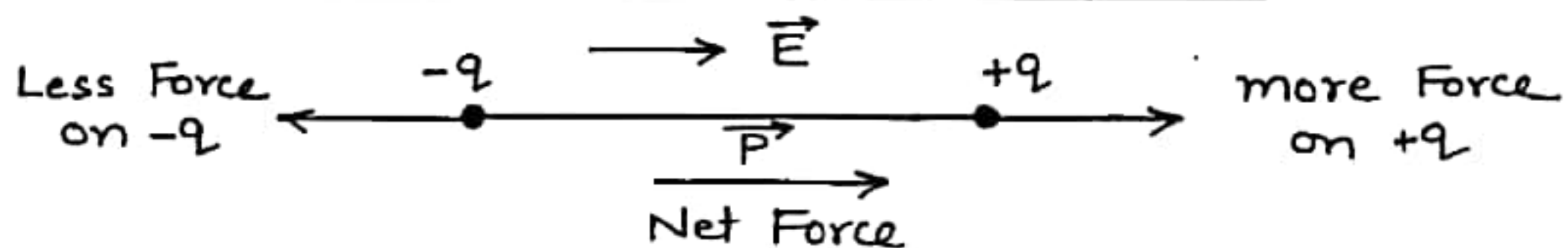
The dipole will be in unstable equilibrium.



Dipole in a non-uniform Electric Field -

In either case, when $\vec{P}$ is parallel to $\vec{E}$ or antiparallel to $\vec{E}$, the net torque is zero, but there is a net force on the dipole if $\vec{E}$ is not uniform.

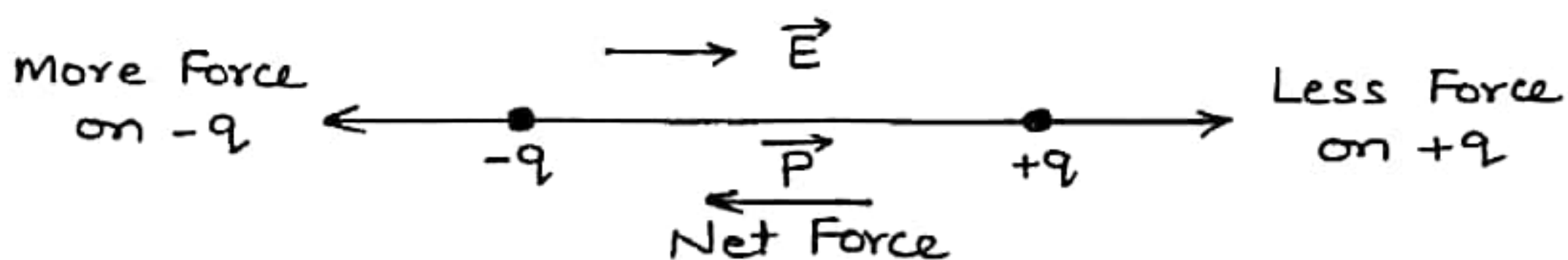
(a) When Field is increasing along $\vec{P}$



Net Force on dipole is along $\vec{E}$.

$$\text{Net Torque} = PE \sin \theta = 0 \quad (\text{as } \theta = 0^\circ)$$

(b) When Field is decreasing along $\vec{P}$



Net Force on dipole is opposite to $\vec{E}$.

$$\text{Net Torque} = PE \sin \theta = 0 \quad \text{as } \theta = 0^\circ$$

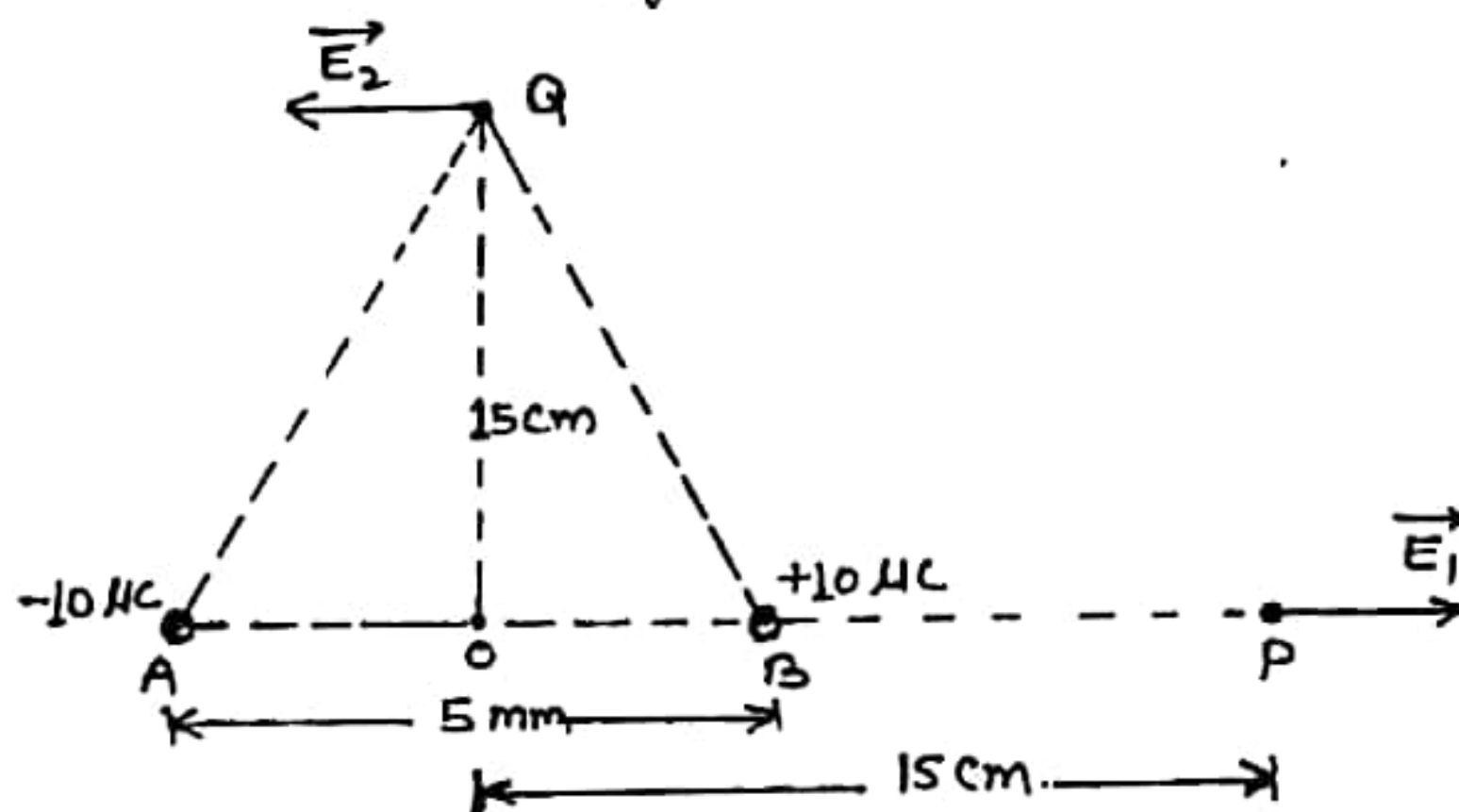
Q. Two charges $\pm 10 \mu\text{C}$ are placed 5 mm apart. Determine the electric field at (a) a point P on the axis of dipole 15 cm away from its centre on the side of the positive charge. (b) a point Q, 15 cm away from O on a line passing through O and normal to the axis of the dipole as shown in figure.

Sol.

Here $|\vec{P}| = q \times 2a$

$$|\vec{P}| = 10 \times 10^{-6} \times 5 \times 10^{-3}$$

$$= 5 \times 10^{-8} \text{ C.m.}$$



(a) As point P lies on axial line of dipole so -

$$E_1 = \frac{2KP}{r^3} = \frac{2 \times 9 \times 10^9 \times 5 \times 10^{-8}}{(0.15)^3}$$

so $E_1 = 2.67 \times 10^5 \text{ N/C}$ (along the direction of $\vec{P}$)

(b) As point Q lies on equatorial line of dipole so -

$$E_2 = \frac{KP}{r^3} = \frac{9 \times 10^9 \times 5 \times 10^{-8}}{(0.15)^3}$$

$E_2 = 1.33 \times 10^5 \text{ N/C}$ (opposite to the direction of $\vec{P}$)

Q. A system has two charges $q_A = 2.5 \times 10^{-7} \text{ C}$ and $q_B = -2.5 \times 10^{-7} \text{ C}$ located at points A: (0, 0, -15 cm) and B: (0, 0, 15 cm) respectively. What are the total charge and electric dipole moment of the system?

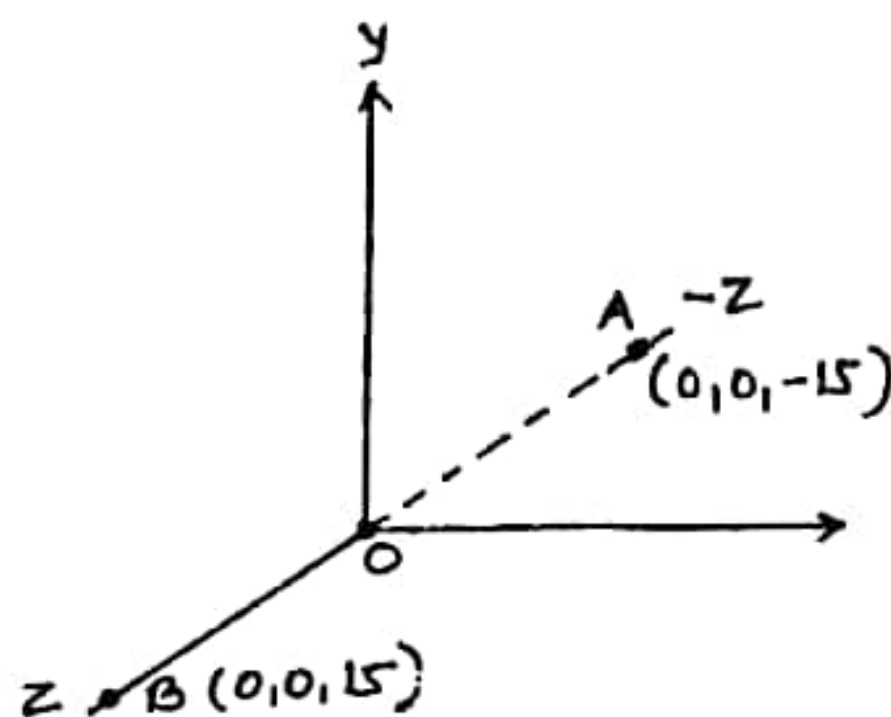
Sol.

Total charge of the system

$$q = q_A + q_B$$

$$q = 2.5 \times 10^{-7} \text{ C} - 2.5 \times 10^{-7} \text{ C}$$

$$q = 0$$



Electric dipole moment of the system is given by-

$$|\vec{P}| = q \times 2a = 2.5 \times 10^{-7} \times 0.30$$

so $|\vec{P}| = 7.5 \times 10^{-8} \text{ C.m (along negative z-axis).}$

[Q.] In a certain region of space, electric field is along the z-direction. The magnitude of electric field is not constant and increases uniformly along positive z-axis, at the rate of 10^5 N/C per metre. Calculate the Force and torque experienced by a system having a total dipole moment 10^{-7} C.m. in the negative z direction?

Sol. Dipole moment of the system $|\vec{P}| = q \times dL$

$$|\vec{P}| = q \times dL = -10^{-7} \text{ C.m.}$$

Rate of increase of electric Field per unit length,

$$\frac{dE}{dL} = 10^5 \text{ N/C per meter}$$

Force experienced by the system is given by

$$F = qE$$

$$\text{or } F = q \times dL \times \frac{dE}{dL}$$

$$F = -10^{-7} \times 10^5 = -10^{-2} \text{ N (along -z axis)}$$

Here the angle between electric field and dipole moment is 180° .

so Torque $\tau = PE \sin \theta$

$$\tau = PE \sin 180^\circ = 0$$

Charge Distribution -

There are two types of distribution of charge on a body -

(1) Discontinuous Charge Distribution -

In this distribution the charges are distributed randomly (non uniformly) in a body at different places.

(2) Continuous charge distribution -

In this distribution the charges are equally distributed in a body or object.

There are three types of continuous charge distribution -

(a) Linear Charge Distribution -

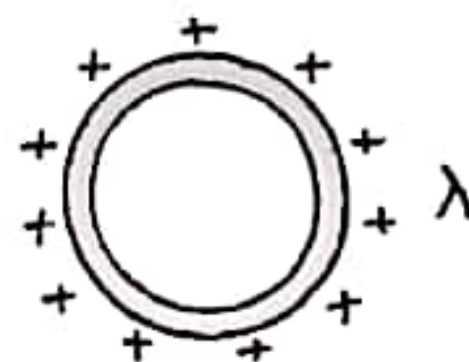
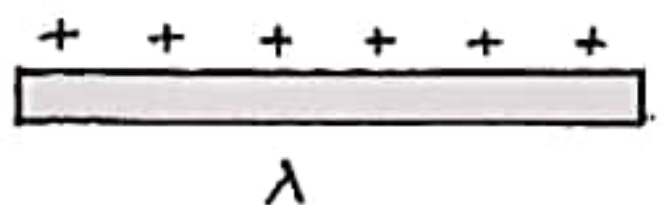
In linear objects the charge is equally linearly distributed on whole body or object.

It is represented by linear charge density (λ).

$$\text{Linear charge density } (\lambda) = \frac{\text{Charge}}{\text{Length}} = \frac{\Delta Q}{\Delta l}$$

* SI Unit $\rightarrow$ C/m.

Examples - Charged straight wire, charged circular ring etc.



(b) Surface charge Distribution

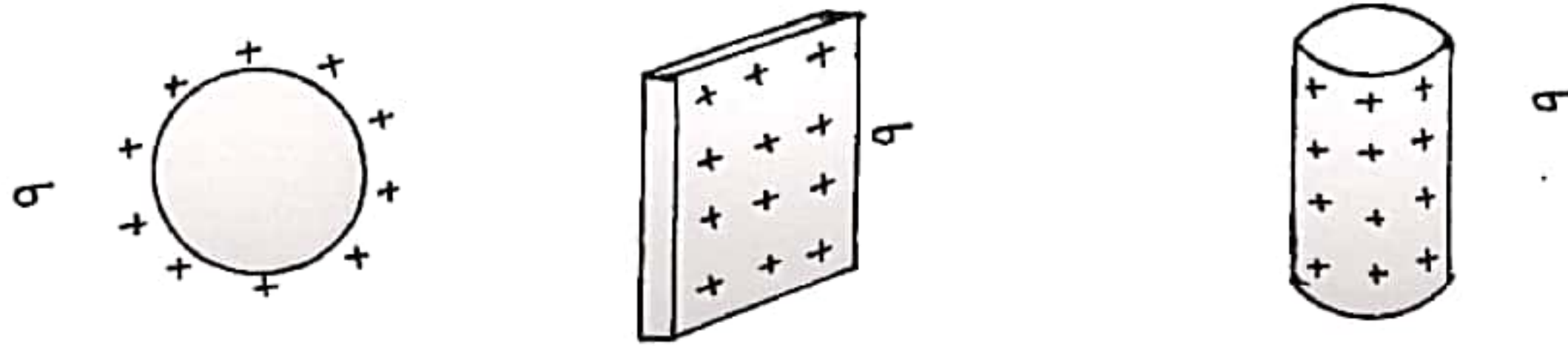
When the surface of the body is uniform in all dimensions then the charge is equally distributed on the whole surface of the body.

It is represented by surface charge density σ .

$$\text{Surface charge density } (\sigma) = \frac{\text{Charge}}{\text{Area}} = \frac{\Delta Q}{\Delta S}$$

SI Unit $\rightarrow C/m^2$.

Example - charged Plane sheet, Charge conducting sphere, cylinder etc.



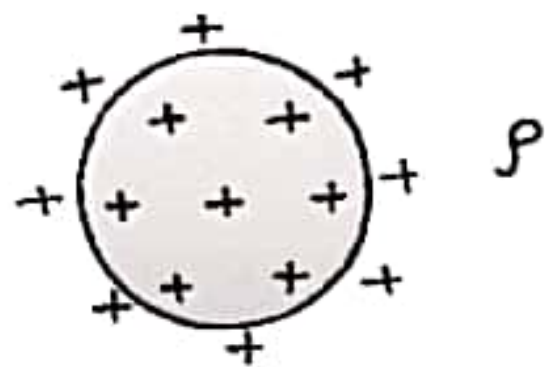
(c) Volume Charge Distribution

When the charge is equally distributed in whole volume of the body then it is represented by volume charge density (ρ).

$$\text{Volume Charge density } (\rho) = \frac{\text{Charge}}{\text{volume}} = \frac{\Delta Q}{\Delta V}$$

SI Unit $\rightarrow C/m^3$.

Examples - charged insulating sphere etc.

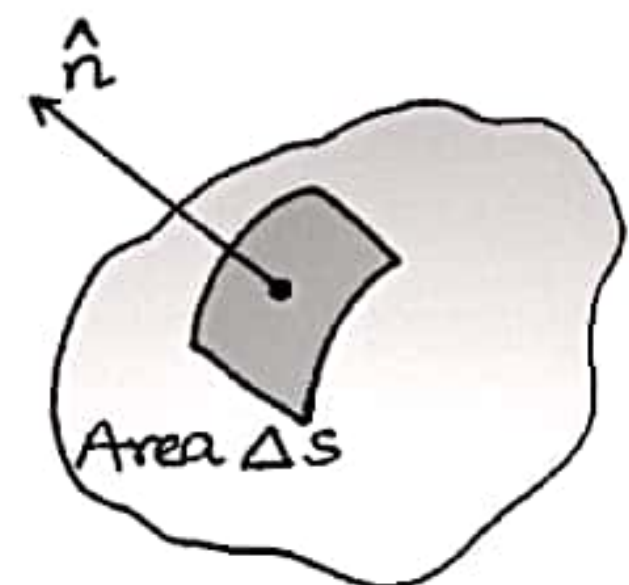


Area Vector -

By convention, the vector associated with every area element of a closed surface is taken to be in the direction of the outward normal.

$$\Delta \vec{S} = (\Delta S) \hat{n}$$

Here ΔS is magnitude of the area element and $\hat{n}$ is a unit vector in the direction of outward normal at that point.



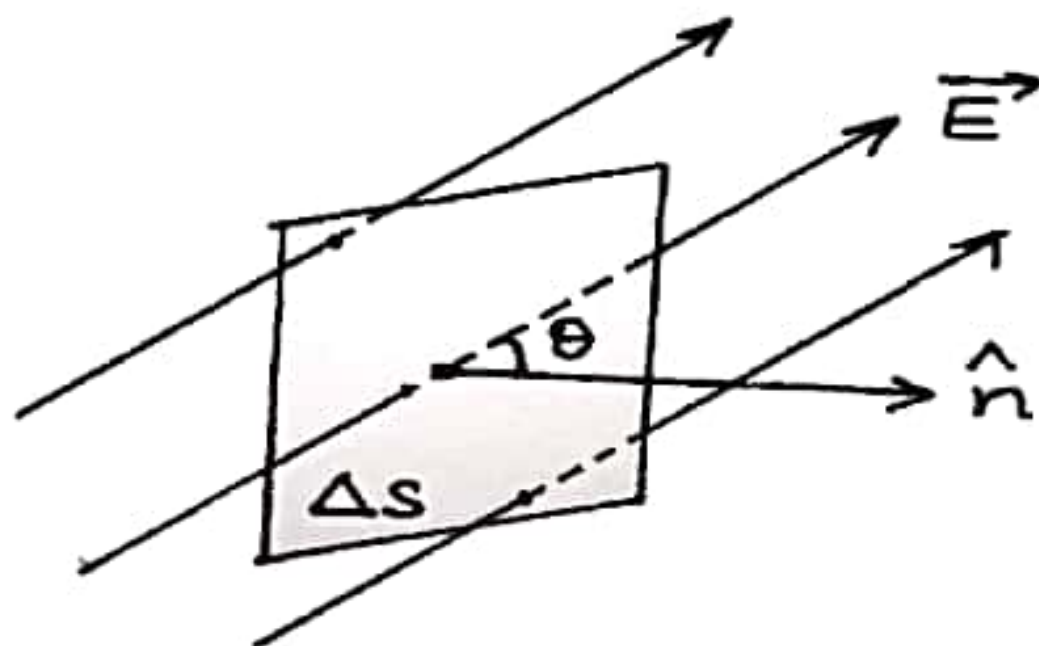
Electric Flux (ϕ) -

Electric flux over an area in an electric field represents the total number of electric field lines crossing this area.

The electric flux $\Delta\phi$ through an area element $\Delta\vec{S}$ in an electric field $\vec{E}$ is defined as -

$$\Delta\phi = \vec{E} \cdot \Delta\vec{S} = E(\Delta S) \cos\theta$$

where θ is the angle between $\vec{E}$ and $\Delta\vec{S}$



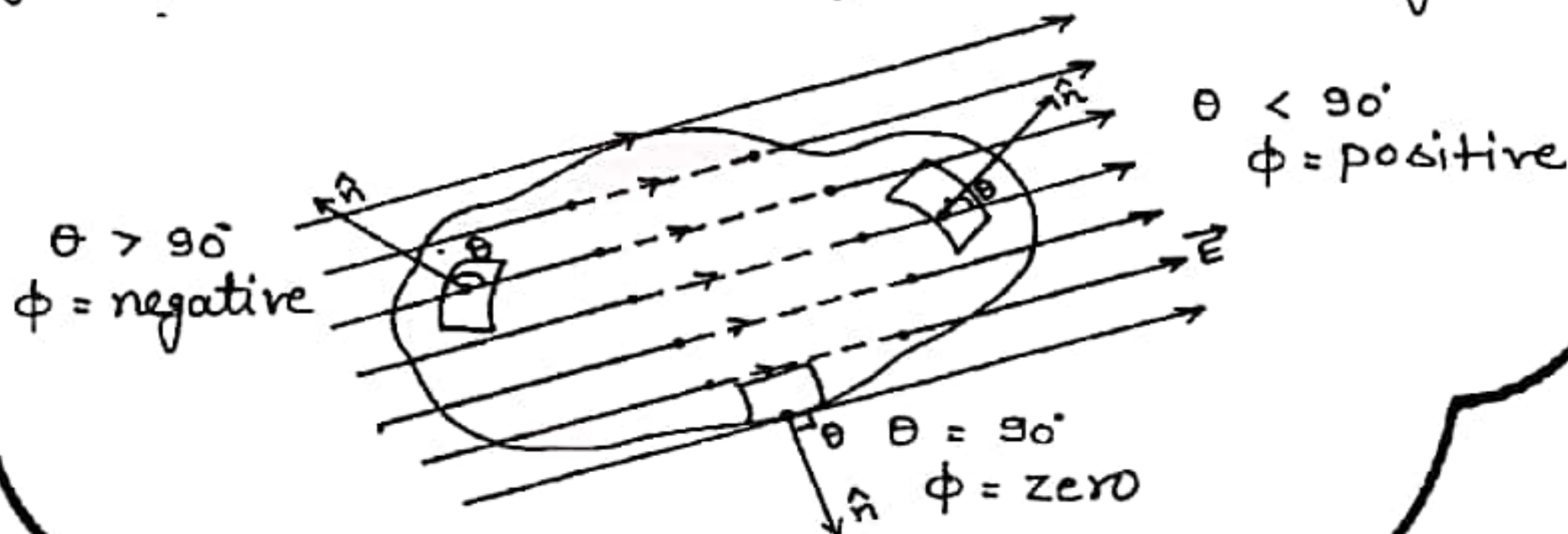
For a non uniform surface total electric flux is

$$\phi = \oint \vec{E} \cdot d\vec{S} = \oint E ds \cos\theta$$

The circle on the integral sign indicates that the surface of integration is a closed surface.

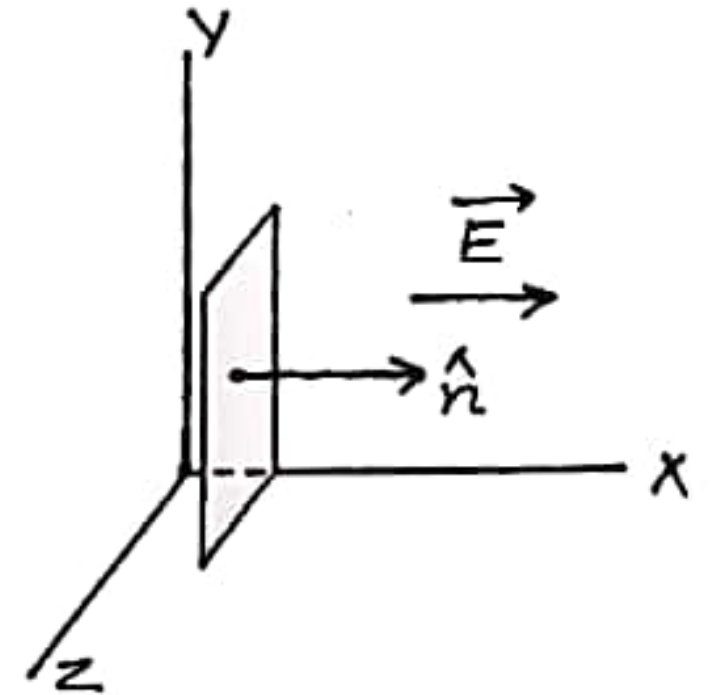
* Electric flux is a scalar quantity.

For a closed surface the flux comes outside from the surface is taken positive and the flux going inside to the surface is taken negative.



Q. Consider a uniform electric field $\vec{E} = 3 \times 10^3 \hat{i} \text{ N/C}$
 (a) What is the flux of this field through a square of 10 cm. on a side whose plane is parallel to the YZ plane? (b) What is the flux through the same square if the normal to its plane makes a 60° angle with the x-axis?

Sol. Here Electric Field $\vec{E} = 3 \times 10^3 \hat{i} \text{ N/C}$
 Side of the square = 0.1 m.
 Area of the square $(A) = (0.1)^2$
 so $A = 0.01 \text{ m}^2$



(a) When the plane of the square is parallel to the Y-Z plane then the angle between the area vector and electric field is $\theta = 0^\circ$.

so Electric Flux $\phi = EA \cos \theta$
 $\phi = 3 \times 10^3 \times 0.01 \times \cos 0^\circ = 30 \text{ Nm}^2/\text{C}$

(b) When plane makes an angle of 60° with the x-axis then $\theta = 60^\circ$.

so Electric flux $\phi = EA \cos \theta$
 $= 3 \times 10^3 \times 0.01 \times \cos 60^\circ$
 $\phi = 15 \text{ Nm}^2/\text{C}$

Q. The electric field in a certain region of space is $(5\hat{i} + 4\hat{j} - 4\hat{k}) \times 10^5 \text{ N/C}$. Calculate electric flux due to this field over an area of $(2\hat{i} - \hat{j}) \times 10^{-2} \text{ m}^2$.

Sol. We know that Electric Flux $\phi = \vec{E} \cdot \vec{ds}$

$$\phi = (5\hat{i} + 4\hat{j} - 4\hat{k}) \times 10^5 \cdot (2\hat{i} - \hat{j}) \times 10^{-2}$$

$$\phi = (10 - 4) \times 10^3$$

$$\phi = 6 \times 10^3 \text{ Nm}^2/\text{C}$$

$$\begin{cases} \hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1 \\ \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0 \end{cases}$$

Gauss's Law -

The total electric flux over the closed surface S in vacuum is $1/\epsilon_0$ times the total charge Q contained inside S .

$$\phi = \oint_S \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0}$$

- * The charges inside S may be point charges or even continuous charge distributions and the location of Q inside S does not affect the value of surface integral.
- * The surface chosen to calculate the surface integral (Electric flux) is called Gaussian surface.

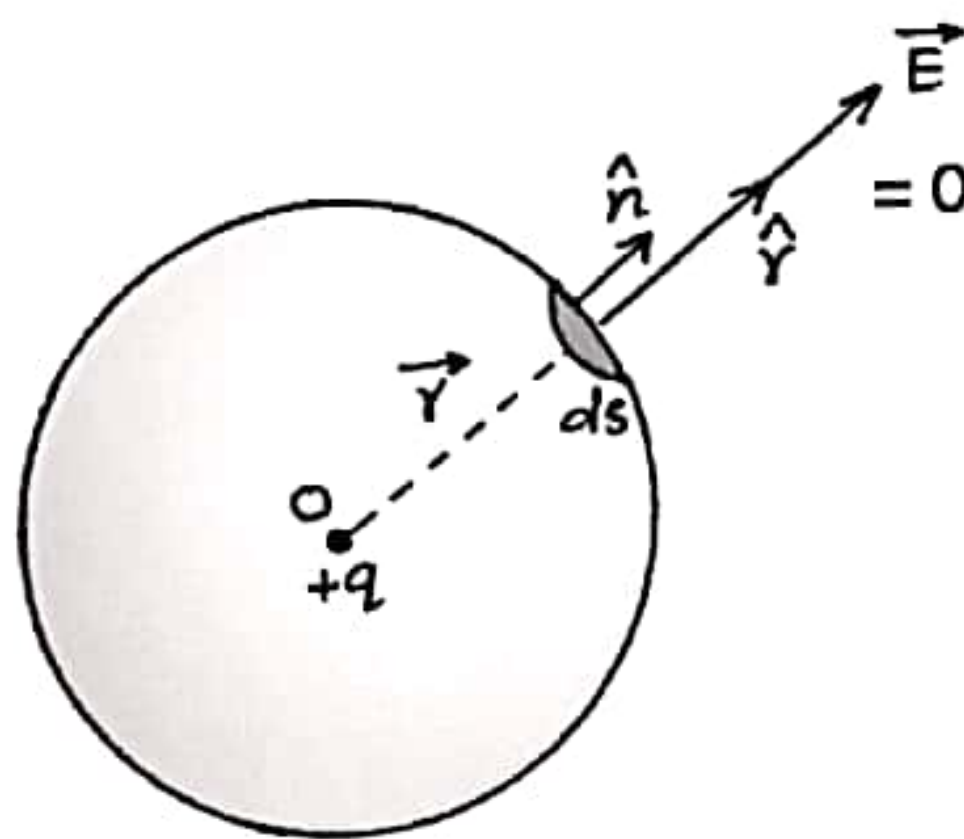
Proof of Gauss's Law -

Suppose an isolated positive point charge q is situated at the centre O of a sphere of radius r .

The electric field intensity at any point P on the surface of the sphere is

$$\vec{E} = \frac{Kq}{r^2} \hat{r}$$

Where $\hat{r}$ is unit vector directed from O to P .



Consider a small area element ds of the sphere around P as -

$$d\vec{s} = ds \hat{n}$$

So the flux over the area element is -

$$d\phi = \vec{E} \cdot d\vec{s} = \frac{Kq}{r^2} \times ds \hat{r} \cdot \hat{n}$$

$$\text{or } d\phi = \frac{Kq}{r^2} ds \quad (\hat{r} \cdot \hat{n} = 1)$$

So Total electric flux over the entire surface of sphere

$$\phi = \oint_S \vec{E} \cdot d\vec{s} = \frac{Kq}{r^2} \oint_S ds$$

$$\phi = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \times 4\pi r^2 \quad \left(\oint_S ds = 4\pi r^2 \right)$$

Hence

$$\phi = \oint_S \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$$

It proves the Gauss's theorem.



If there are n point charges $q_1, q_2, \dots, q_n$ inside the closed surface then -

$$\phi = \frac{Q}{\epsilon_0}$$

(where $Q = q_1 + q_2 + \dots + q_n$)

Q. A charge of $2 \times 10^{-9} \text{ C}$ is placed on a corner of a cube of side 1 m . Find the electric flux passing through this cube.

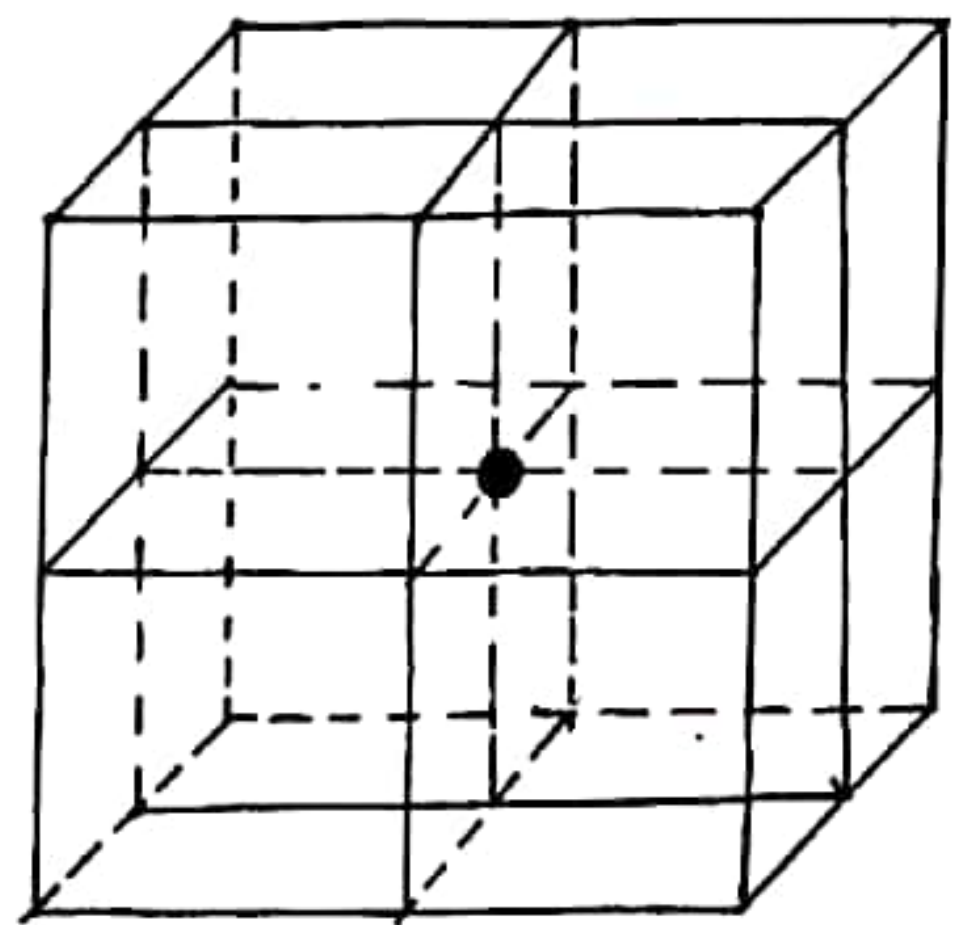
Sol.

Here eight cubes will be required to form a Gaussian surface such that the given charge of $2 \times 10^{-9} \text{ C}$ appears at the centre of the bigger cube.

Thus flux $\phi = \frac{1}{8} \times \frac{q}{\epsilon_0}$

$$\phi = \frac{1}{8} \times \frac{2 \times 10^{-9}}{8.85 \times 10^{-12}}$$

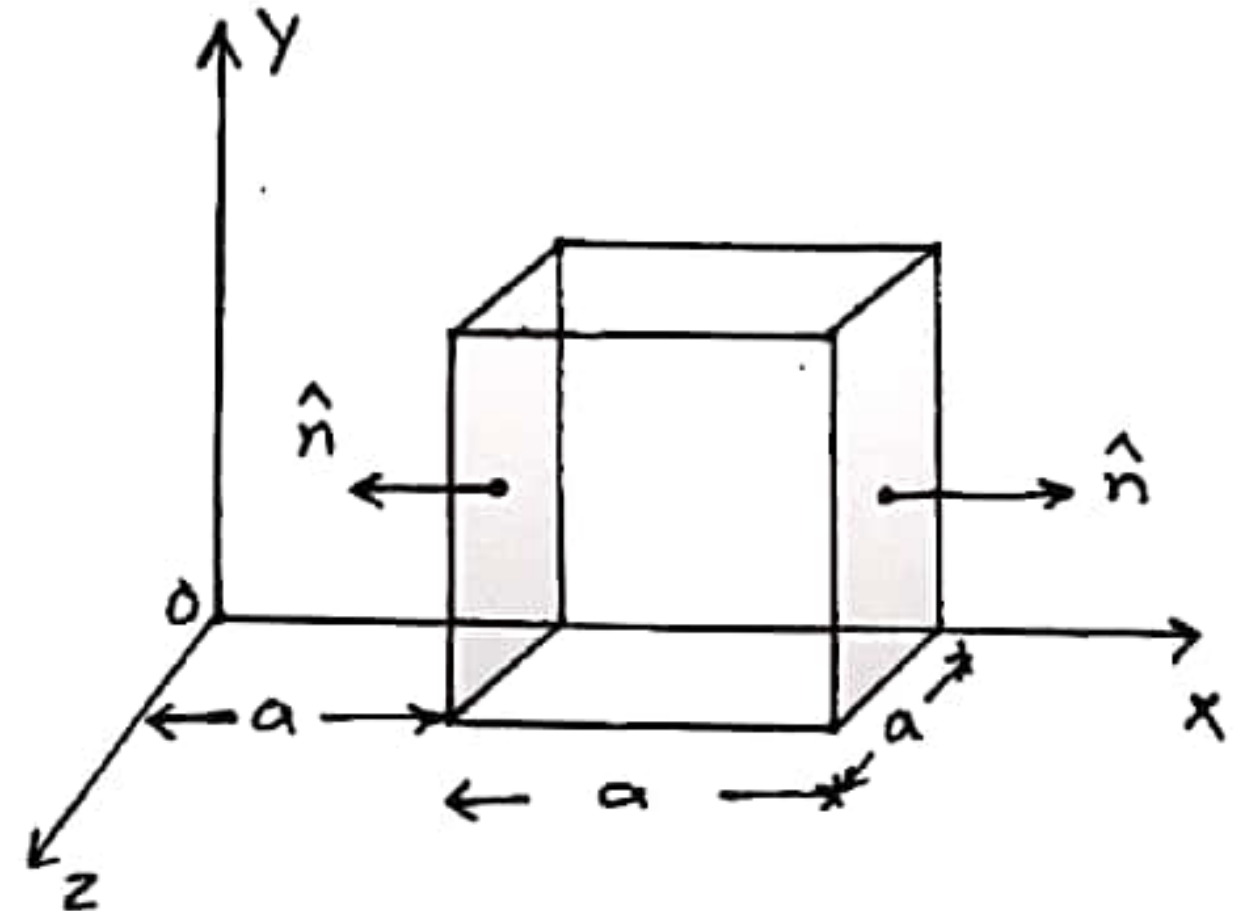
$$\phi = 28.25 \text{ Nm}^2/\text{C}$$



- Q.** The electric field components in fig. are $E_x = \alpha x^{1/2}$, $E_y = E_z = 0$ in which $\alpha = 800 \text{ N/C-m}^{1/2}$. Consider the cube as shown in fig and calculate -
 (a) Total flux through the cube
 (b) Charge within the cube. Assume that $a = 0.1 \text{ m}$.

Sol. Here $E_x = \alpha x^{1/2}$
 $E_y = E_z = 0$,
 $\alpha = 800 \text{ N/C-m}^{1/2}$
 and $a = 0.1 \text{ m}$.

(a) The electric field has only x component, so
 $\phi = \vec{E} \cdot \vec{dS} = 0$ for each of four faces of cube perpendicular to $Y-Z$ plane.



At the left face $x = a$ so $E_L = \alpha a^{1/2}$

so $\phi_L = \vec{E}_L \cdot \vec{\Delta S} = \alpha a^{1/2} \times a^2 \cos 180^\circ = -\alpha a^{5/2}$

At the Right Face $x = a + a = 2a$ so $E_R = \alpha (2a)^{1/2}$

so $\phi_R = \vec{E}_R \cdot \vec{\Delta S} = \alpha (2a)^{1/2} \times a^2 \cos 0^\circ$

$$\phi_R = \alpha a^{5/2} \sqrt{2}$$

So Net flux through the cube $\phi = \phi_R + \phi_L$

$$\phi = \alpha a^{5/2} \sqrt{2} - \alpha a^{5/2} = \alpha a^{5/2} (\sqrt{2} - 1)$$

$$\phi = 800 (0.1)^{5/2} (\sqrt{2} - 1) = 1.05 \text{ Nm}^2/\text{C}.$$

(b) By Gauss's theorem

$$q = \epsilon_0 \phi$$

so $q = 8.85 \times 10^{-12} \times 1.05$

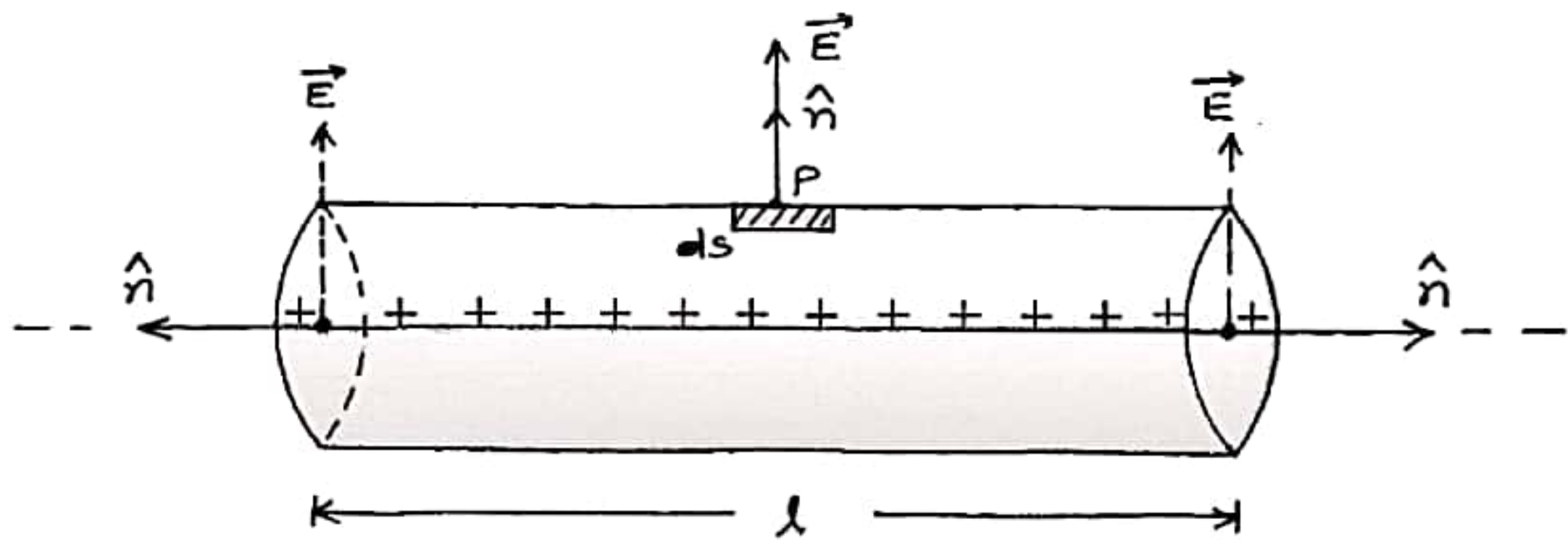
$$q = 9.27 \times 10^{-12} \text{ C}.$$

Application of Gauss's Theorem

(1) Field due to an infinitely Long straight Uniformly charged wire -

Consider an infinitely long thin wire with uniform linear charge density λ .

The electric field at every point in the plane cutting the wire normally is radial and its magnitude depends only on the radial distance r .



Now consider a circular closed cylinder of radius r and length l , with the infinitely long line of charge as its axis.

On the end of cylinder, angle between electric field intensity is $\vec{E}$ and unit vector $\hat{n}$ is $\theta = 90^\circ$. So the flux related to these circular portions will be zero.

The electric field at every point on the curved surface of the cylinder is the same. Also the $\vec{E}$ and unit vector $\hat{n}$ are along outward normal to the curved surface so $\theta = 0^\circ$.

So the electric flux over the curved surface -

$$\phi = \oint_s \vec{E} \cdot d\vec{s} = \oint_s E ds \cos 0^\circ$$

so

$$\boxed{\phi = E \oint_s ds = E \times 2\pi r l}$$

Charge inside the cylinder (q) = λl

Now according to Gauss's theorem -

$$\phi = \frac{q}{\epsilon_0}$$

so $E \times 2\pi r l = \frac{\lambda l}{\epsilon_0}$

or $\vec{E} = \frac{\lambda}{2\pi \epsilon_0 r} \hat{n}$ clearly $E \propto \frac{1}{r}$



- * If $\lambda > 0$, the direction of electric field at every point is radially outwards.
- * If $\lambda < 0$, the direction of electric field at every point is radially inwards.

Q. A plastic rod of length 2.2 m and radius 3.6 mm carries a negative charge of $3.8 \times 10^{-7} \text{ C}$ which is spread uniformly over its surface. What is the electric field near the mid point of the rod, on its surface?

Sol: Given that - $l = 2.2 \text{ m}$.
 $r = 3.6 \text{ mm} = 3.6 \times 10^{-3} \text{ m}$, $q = -3.8 \times 10^{-7} \text{ C}$.

Linear charge density $\lambda = \frac{q}{l}$

$$\lambda = \frac{-3.8 \times 10^{-7}}{2.2} = -1.73 \times 10^{-7} \text{ C/m}$$

As $E = \frac{\lambda}{2\pi \epsilon_0 r} = \frac{2\lambda}{4\pi \epsilon_0 r}$

so $E = \frac{2 \times 9 \times 10^9 \times (-1.73 \times 10^{-7})}{3.6 \times 10^{-3}}$

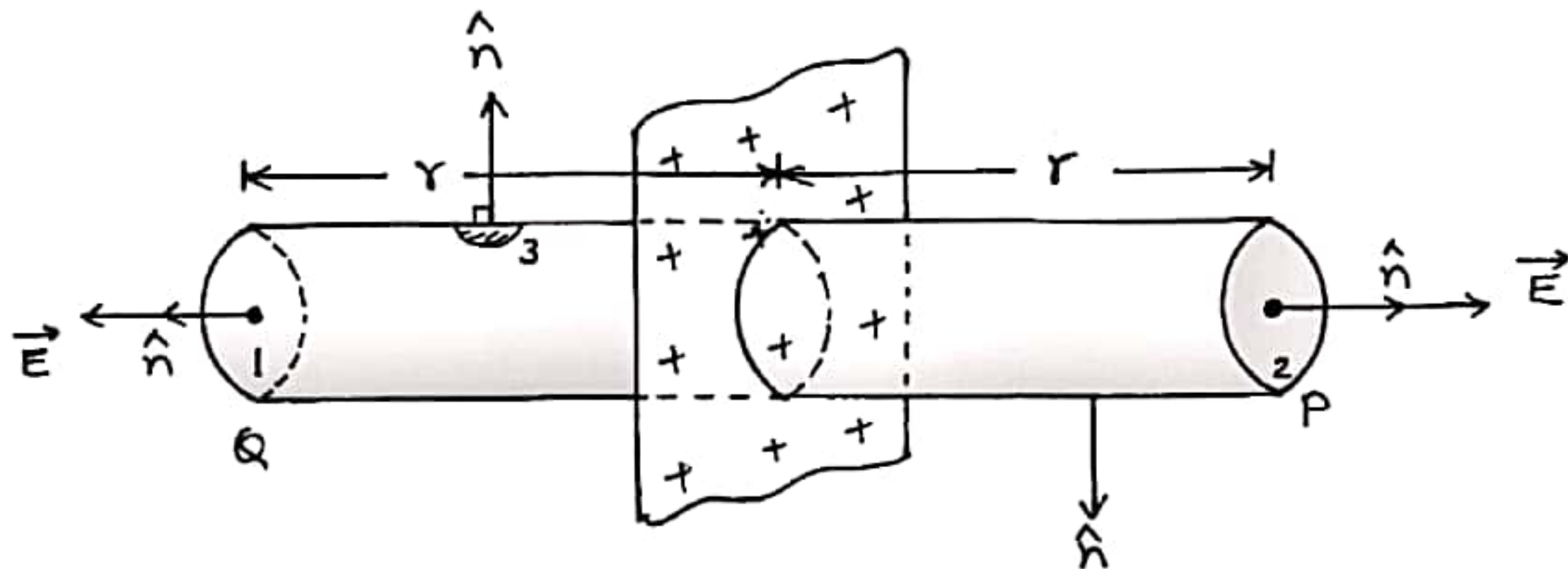
or $E = -8.6 \times 10^5 \text{ N/C}$

(3) Electric Field intensity due to a thin infinite plane sheet of charge -

Consider a thin infinite plane sheet of charge with surface charge density σ .

We have to find the electric field at point P at a perpendicular distance r from the sheet.

Now imagine a cylindrical Gaussian surface of cross-sectional area A around P and length $2r$, passing through the sheet.



Total flux over the entire surface of cylinder is -

$$\phi = \int_1 \vec{E} \cdot d\vec{s} + \int_2 \vec{E} \cdot d\vec{s} + \int_3 \vec{E} \cdot d\vec{s}$$

$$\phi = E ds \cos 0^\circ + E ds \cos 0^\circ + E ds \cos 90^\circ$$

$$\phi = 2EA$$

Total Charge enclosed by the cylinder = σA

According to Gauss's Law -

$$\phi = \frac{q}{\epsilon_0} \Rightarrow 2EA = \frac{\sigma A}{\epsilon_0}$$

So

$$E = \frac{\sigma}{2\epsilon_0}$$

* E is independent of r .

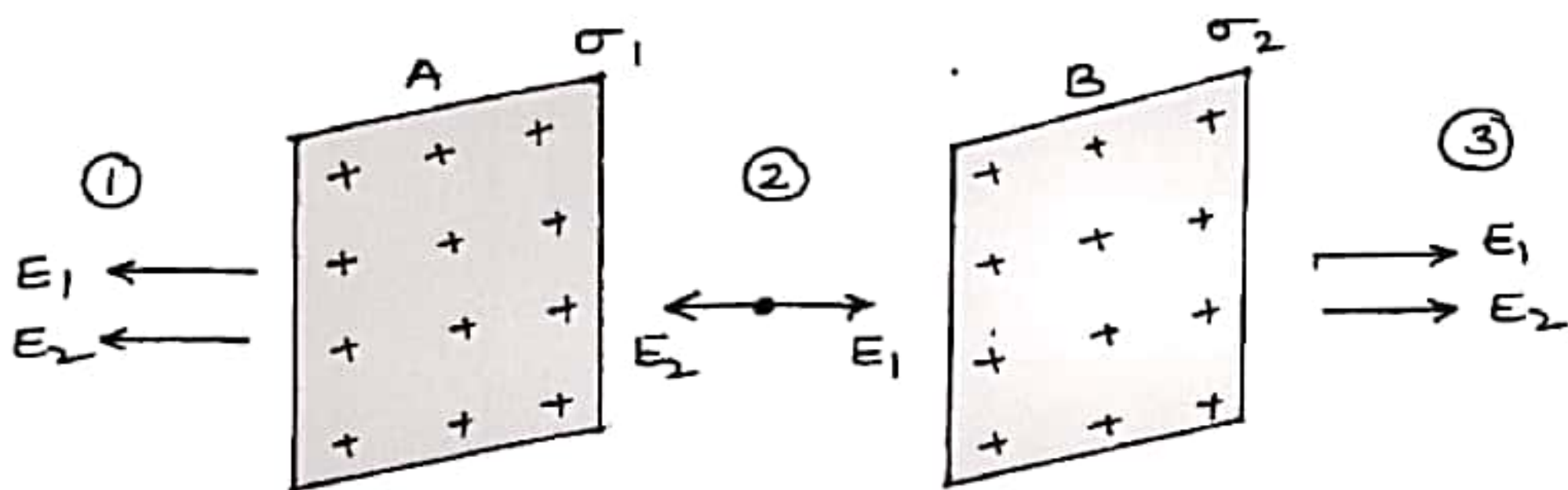
* If $\sigma > 0$ then electric field is normally outward and

* If $\sigma < 0$ then electric field is normally inward to the sheet.

(4) Electric Field intensity due to two thin infinite parallel sheets of charge -

Case 1: -

Here $\sigma_1 > \sigma_2 > 0$



In region ① & ③

$$E = E_1 + E_2$$

$$E = \frac{\sigma_1}{2\epsilon_0} + \frac{\sigma_2}{2\epsilon_0}$$

$$E = \frac{1}{2\epsilon_0} (\sigma_1 + \sigma_2)$$

In region ②

$$E = E_1 - E_2$$

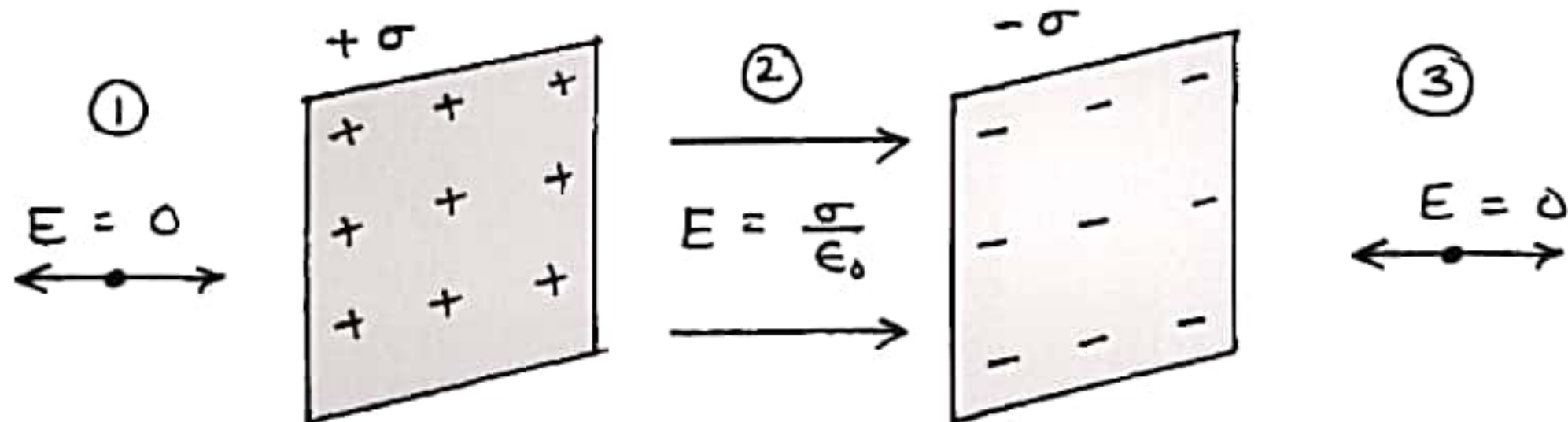
$$E = \frac{\sigma_1}{2\epsilon_0} - \frac{\sigma_2}{2\epsilon_0}$$

$$E = \frac{1}{2\epsilon_0} (\sigma_1 - \sigma_2)$$

Case 2: -

Here $\sigma_1 = \sigma$

and $\sigma_2 = -\sigma$



In region ① & ③

$$E = E_1 - E_2$$

$$E = \frac{\sigma}{2\epsilon_0} - \frac{\sigma}{2\epsilon_0}$$

$$E = 0$$

In region ②

$$E = E_1 + E_2$$

$$E = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0}$$

$$E = \frac{\sigma}{\epsilon_0}$$

Q. A Large plane sheet of charge having surface charge density $5 \times 10^{-6} \text{ C/m}^2$ lie in xy plane. Find the electric flux through a circular area of radius 0.1 m if the normal to the circular area makes an angle of 60° with the z-axis.

Sol. Here $\sigma = 5 \times 10^{-6} \text{ C/m}^2$, $r = 0.1 \text{ m}$, $\theta = 60^\circ$

$$\text{So } \phi = \vec{E} \cdot \vec{ds} = E ds \cos \theta$$

$$\phi = \left(\frac{\sigma}{2\epsilon_0} \right) \times \pi r^2 \cos \theta$$

$$\phi = \frac{5 \times 10^{-6}}{2 \times 8.85 \times 10^{-12}} \times 3.14 \times (0.1)^2 \cos 60^\circ$$

$$\phi = 4.44 \times 10^3 \text{ Nm}^2/\text{C} \quad \text{--- Ans.}$$